



SAPIENZA
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Human Computer Interaction

ESPark

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Sommario

Introduction	2
1 Requirement analysis	3
1.1 Competitors analysis	3
1.2 Questionnaire analysis.....	5
1.3 Users analysis.....	12
1.3.1 User profile.....	12
1.3.2 Personas and scenarios	12
2 Task and Dialogue analysis: HTA & STN	15
2.1 Hierarchical Task Analysis	15
2.1.1 Look for an available parking slot.....	15
2.1.2 Pay the parking on the app	16
2.1.3 Extend parking duration	17
2.2 State Transition Networks	18
2.2.1 Look for an available parking slot.....	18
2.2.2 Pay the parking on the app	18
2.2.3 Extend parking duration.....	19
3 Mockup: prototype 1	20
4 Expert Based Evaluation	25
4.1 Heuristic Evaluation	25
4.2 Cognitive Walkthrough.....	29
4.2.1 Tasks analyzed	29
4.3 Prototype 2.....	38
5 User Based Evaluation	45
5.1 Think Aloud	45
5.2 Controlled Experiment	46
5.2.1 Experiment 1	46
5.2.2 Experiment 2	51
6 Implementation and Final Product.....	55
6.1 Hardware implementation.....	55
6.2 Software implementation	56
7 Conclusions and Future Work	62
7.1 Conclusions	62
7.2 Future Work	62

Introduction

The project concerns the creation of an app to facilitate the user's search for an available parking slot. The main goal of the app is to make users avoid endless searches for a parking slot; by using our app the user can first view which parking space is available in real-time, in particular he/she will be able to locate the available parking slot in certain sectors or floors of the parking lot, by doing so he/she will be able to reach the available parking slot optimizing the search. The app offers the user the possibility to select the desired parking slot before his arrival; by doing so, the user will receive a notification if the parking slot is no longer available. In the event that parking is subject to payment, the user can pay by credit card and has the option of extending the duration of your stay via the app.

A practical example to use our app can be for example in parking lots in shopping malls where there are also multi-level parking lots, the user therefore can view on the app what floor a parking space is free and directly reach the said floor optimizing the time.

The idea, therefore, is to make the app by going to simulate the parking lots with an ESP32 microcontroller that, through different sensors (which simulate the parking slot), allows to simulate the availability or not of the parking slot, so that app users can see in real time whether parking slots are available or not.

Our aim is to use technology to help people in their daily lives and above all in those actions that would seem obvious but have a great impact on many aspects of people's lives and our society: quality of life, environmental issues (in terms of consumption in continuing to look for a parking space), and time that we can use for other activities.

Chapter 1

Requirement analysis

1.1 Competitors analysis

For this type of project there are no apps that implement the same idea, but an analysis was made on apps that offer a similar service on finding parking that this project offers. In general, current parking apps have the main focus of allowing payment via smartphone; instead the main purpose of this project is to avoid endless searches for a parking space by viewing the free parking slots in the car parks on the app. The following competitor apps were found:

- EasyPark
- Parkopedia
- Telepass

EasyPark: is the app that comes closest to the idea of this project. The main purpose of the app is to allow users to pay for parking directly from their smartphone. Furthermore, through the FIND function it allows you to view the percentage of occupancy of the parking lots. The aforementioned features contain the advantages of the app; but the app also has some disadvantages which are as follows: it forces the user to insert the card even for free parking and communicates the percentage of occupancy of the parking spaces only with respect to the vehicles whose drivers use the app.

Parkopedia: is another application for parking. To search for a car park, simply select "Current position" to search nearby, or enter a street or city in the search field above. Thus Parkopedia will show the various results in a map with colored icons according to the price. One of the strengths of Parkopedia lies precisely in the search function, which can be managed on the basis of various parameters. For this there are filters with which to skim everything based on the time of entry, the duration of the stay, the type (garage, street, etc.), the payment methods, the presence of charging stations for electric cars and so on. This app is only used to search for parking based on filters, in fact it does not allow payment and does not provide any information on parking availability.

Telepass: nother possible option is to use the TELEPASS application to search for a parking space, but the latter can only be used in the parking lots of airports, stations and fairs with which agreements have an agreement. In particular this app is useful to pay the blue line parking, and it is not a really application to look for an available parking slot. With the device you will be able to enter and exit the car parks in the structure in the reserved lane without buying a ticket, looking for the column with the Telepass logo, the

bar will rise automatically and you will be able to leave again without stopping to pay at the cash desk.

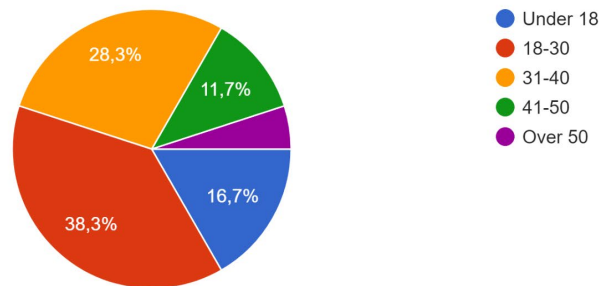
	ESPark	EasyPark	Parkopedia	Telepass
Account (profile, vehicles owned, payment methods)				
Map to choose the parking				
Filters to choose the suitable car park				
Pay for parking with the app				
Extend your parking duration with the app				
Real-time update of the availability or non-availability of a parking slot		 (only a percentage estimate of the availability of the entire car park)		

1.2 Questionnaire analysis

This paragraph reports the results of the questionnaire created with "Google Forms". As much information as possible of a different nature was collected, to better understand potential targets for our app and also to understand the enthusiasm of users for our idea in order to improve or confirm several initial ideas for the app.

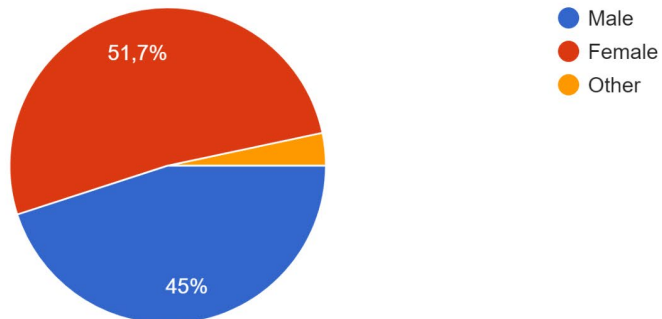
Age

60 risposte



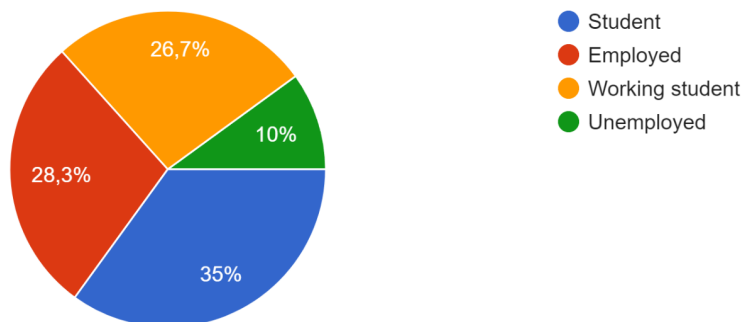
Gender

60 risposte



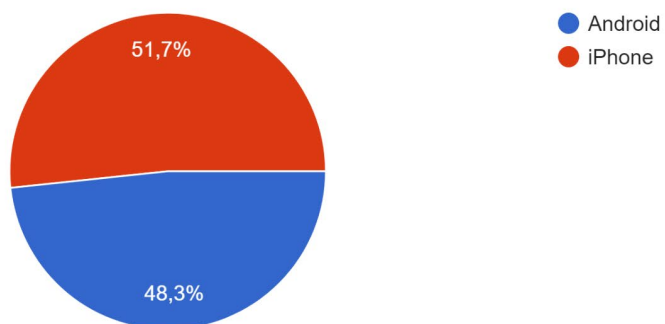
What is your current status?

60 risposte



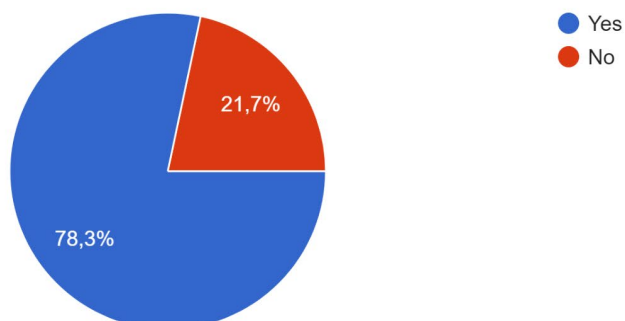
Type of smartphone?

60 risposte



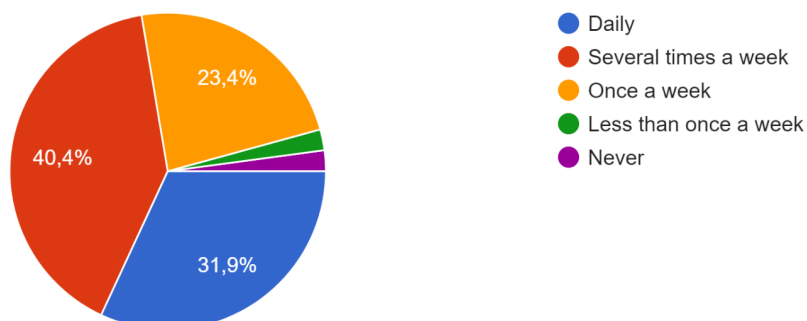
Do you have a driving licence? (of any type, e.g. AM, B, A, ...)

60 risposte



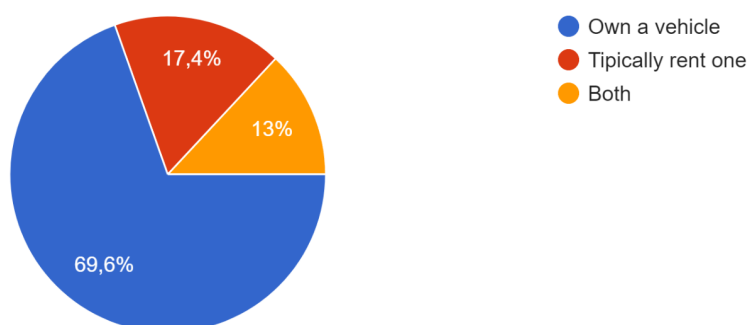
How frequently do you drive to your destination? (ex. to university, to work, to cinema, ...)

47 risposte



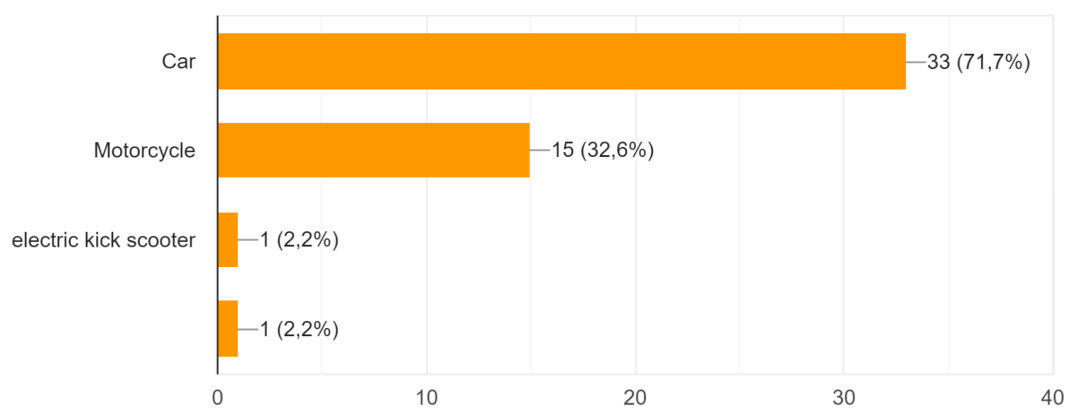
Do you have your own vehicle or do you rent one?

46 risposte



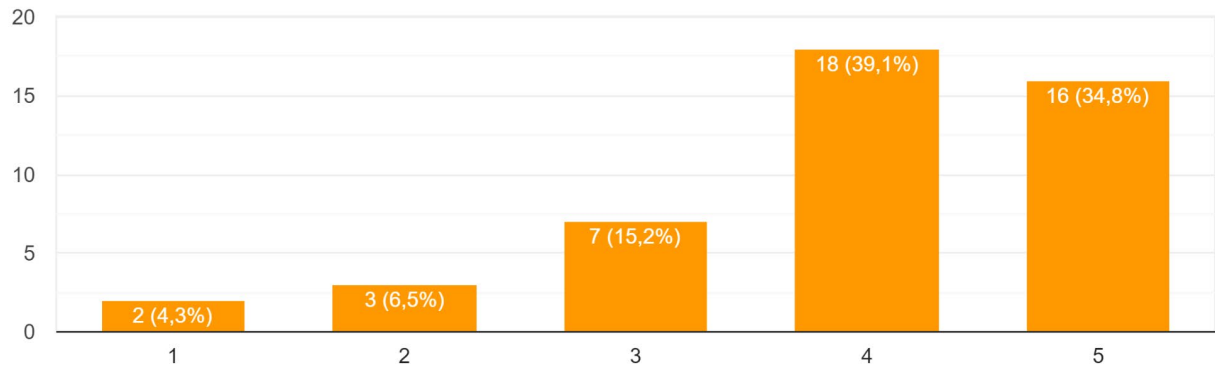
What type of vehicle do you use to reach your destination? (you can select more than one)

46 risposte



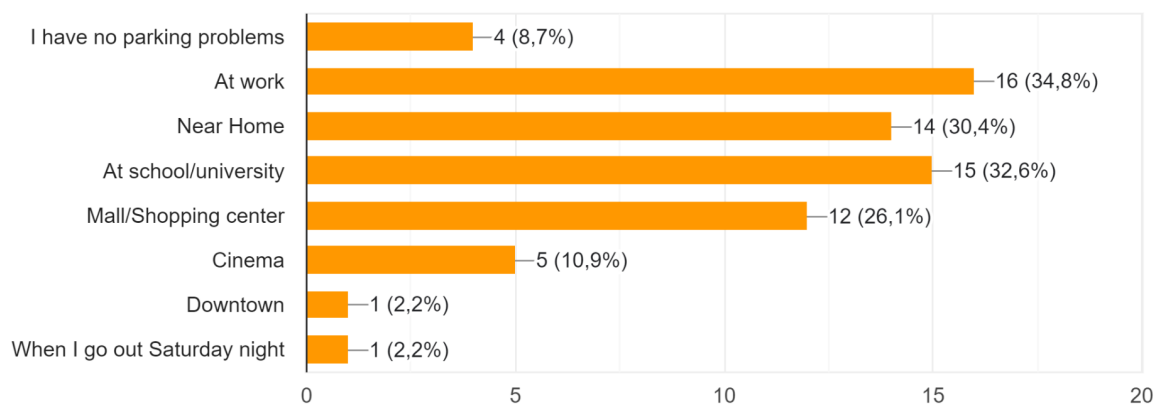
How often do you struggle to find parking in the area you're driving to?

46 risposte



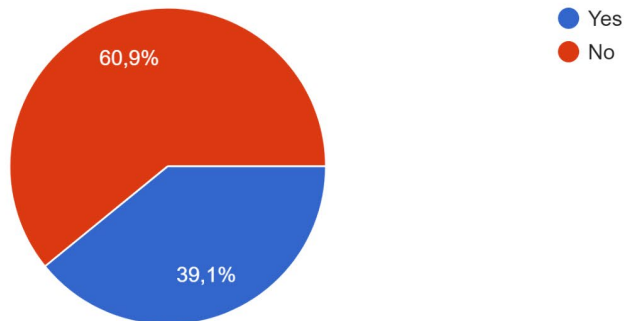
Where do you find it most difficult to find parking? (you can select more than one)

46 risposte



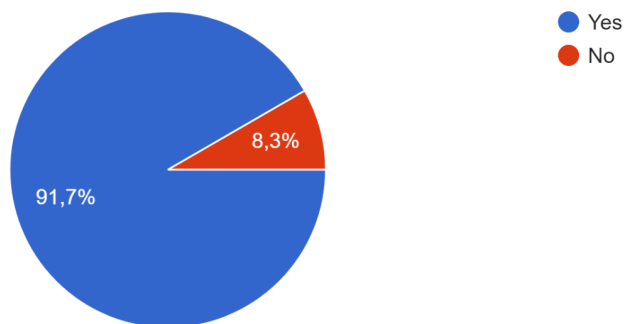
Do you use a parking app?

46 risposte



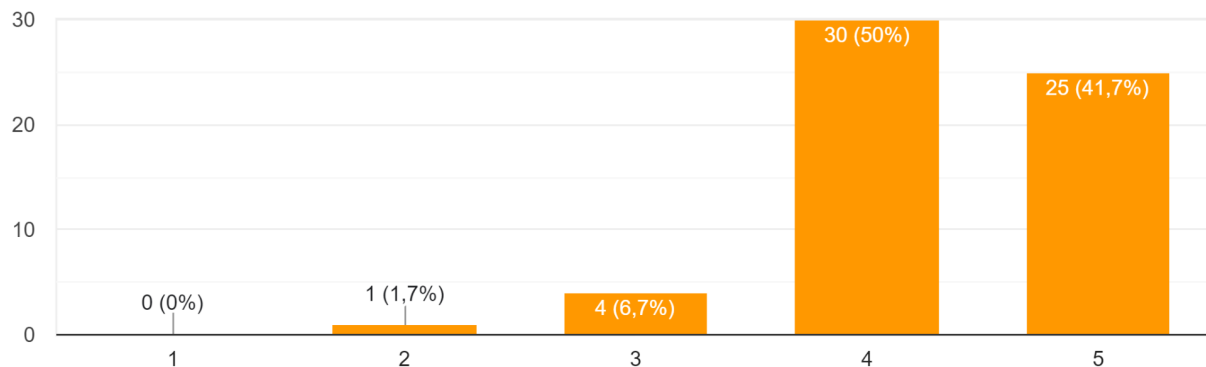
Do you find an app that shows the availability of parking space in real-time interesting?

60 risposte



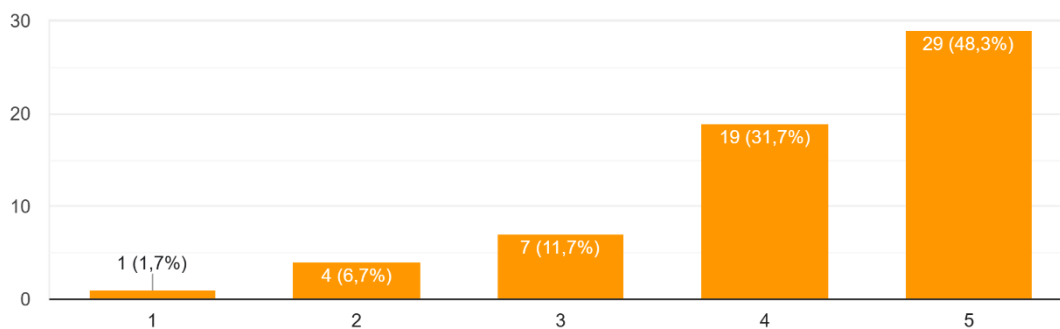
How useful do you think it would be to display the entire parking lot with free and/or occupied spaces, by marking them in red or green?

60 risposte



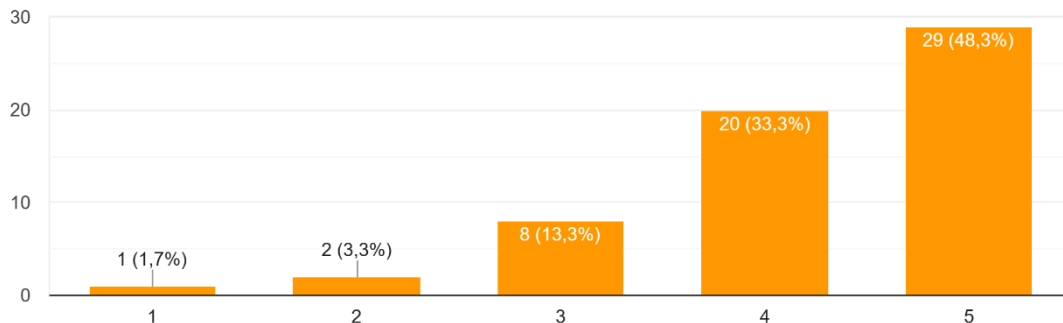
How useful do you think it is for the user to pay for parking directly from the app?

60 risposte



How useful do you think it is for the user to extend the parking duration directly from the app, without returning to the vehicle?

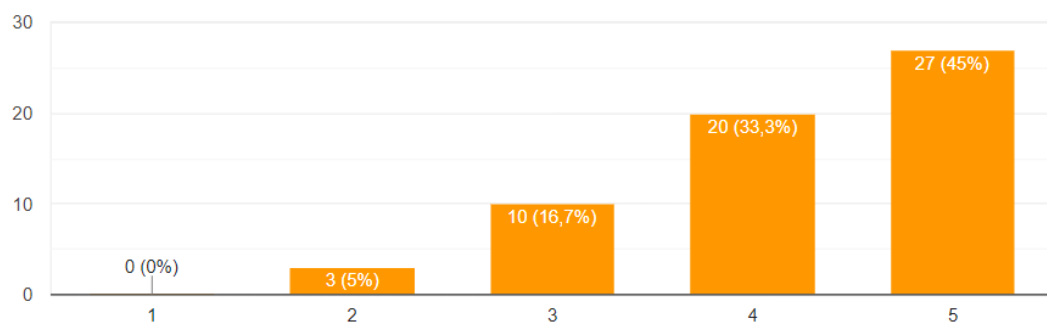
60 risposte



How useful do you think it is for the user to choose an available parking slot before arriving based on need (i.e. closest to the entrance, closest to a shop etc), so that the app can notify the user whether the slot remains available or is occupied before the user arrives?



60 risposte



Conclusions

From the analysis of the questionnaire we obtained interesting information.

Most users use a vehicle to reach their destinations several times a week, typically using a car or motorbike. Most users have a lot of difficulty finding parking, in various places such as: at work, at university, during their free time, etc. About 60% of users do not use parking apps, but about 91% of users are interested in using our app to see the availability of a parking slot in real time. Finally, we asked users how useful they find different features of our applications, in most cases users were interested in using these features in our app.

1.3 Users analysis

This app is aimed at all users who have a driving license of any type and who use the car/motorcycle to get around, i.e. people aged 14 and over.

1.3.1 User profile

- **Age:** 14-65 years
- **Gender:** Any
- **Occupation:** Any
- **Location:** Any
- **Technology:** smartphone with GPS and internet connection
- **Vehicle:** car/motorcycle

1.3.2 Personas and scenarios

Now several imaginary people have been analyzed who identify the typical user who is interested in using this app in everyday life and moreover scenarios have been associated with each person.

Persona 1: Giulia



Giulia is a 29 years old commercial advisor born and grown in Roma in Garbatella district. She is a girl full of energy, so she loves to make every kind of activities. As soon as she started working, Giulia decided to buy a new car to facilitate her travels, not only for work but also to facilitate the activities she loves after work, such as theatre classes, but also going to the cinema with friends, and tennis matches with work colleagues. In Rome she lives in a house with other two girls, one a lawyer and one a university student of design. Until she started working, she never used the car, so she used public transport to get around. However, since using the car, Giulia has been thinking of moving to a smaller city, as the problem of parking in such a large city has become very stressful for all her activities, every morning when she must go to work Giulia spends 30 minutes to look for a free parking space, plus another 30 when she must look for it as close to home

as possible. Not to mention when he goes to visit his aunt Carla in via Tuscolana, it becomes very stressful.

Scenario:

Like every Sunday Giulia decides to go to visit Aunt Carla in via Tuscolana, to get there from Garbatella it will take more or less 20 minutes by car, and she knows very well that it is very difficult to find parking near her aunt's house, so she relies on a new application to book a parking space, before getting into the car, through your smartphone, book a parking space among the free ones available, directly from the application, so that once you arrive near your aunt's home, you won't have to waste time looking parking and spending more time with the aunt.

Persona 2: Carl



Carl is a 20-year-old Kenyan boy, a chemical engineering student who moved to Rome thanks to a scholarship. He loves Italy and has always dreamed of living there one day, but he didn't expect to find himself faced with enormous difficulties in reaching the venue for his lessons, also due to the distance from his home to the university headquarters, given the different cost of life compared to his country, to spend less on rent he decided to take a room with other university students in a more distant area. He therefore decides to buy a small-engine motorcycle to facilitate his travels, but even in this case he wastes a lot of time every day, often being late for lessons to look for a parking space near the engineering headquarters.

Scenario:

It is Monday, lessons start at 9:00 AM and Carl just got out from his house at 8.30, Carl being aware of the traffic in the city of Rome, and of the problems in finding parking that he would have encountered once he arrived near the university, decides to rely on an application in such a way that once he arrives at the university he has the possibility to view directly the free parking available closest to him, so as not to miss even a minute of the lesson.

Persona 3: Maria



Maria is 33 years old interior designer from Florence, who recently moved to Rome. Not having a stable job, she decided not to invest in buying a car given the high costs for the purchase but also for the high management costs, therefore for all the movements she uses public transport. Maria, being an interior designer, has a great passion to go to art exhibitions but her biggest passion is to furnish her room to her liking, so her favourite outing is to go to Ikea for shopping.

Scenario:

Maria decides to go with her work colleagues this Sunday to Ikea, but since Ikea is not well connected by public transport, she decides to rent a car, not being an expert in driving using the car very little, she already knows who will have trouble looking for parking, and would like to avoid making very complex parking, so she uses her smartphone to find an available parking space inside the shopping centre that is free for her arrival, moreover because a friend of hers has a motor disability, thanks to the application, he can find a parking space that can make it easier for his colleague to get out of the car near the entrance to the shopping centre. All this without having to waste time looking for parking and spending that time looking at how to spruce up her room and spend time with her new friends.

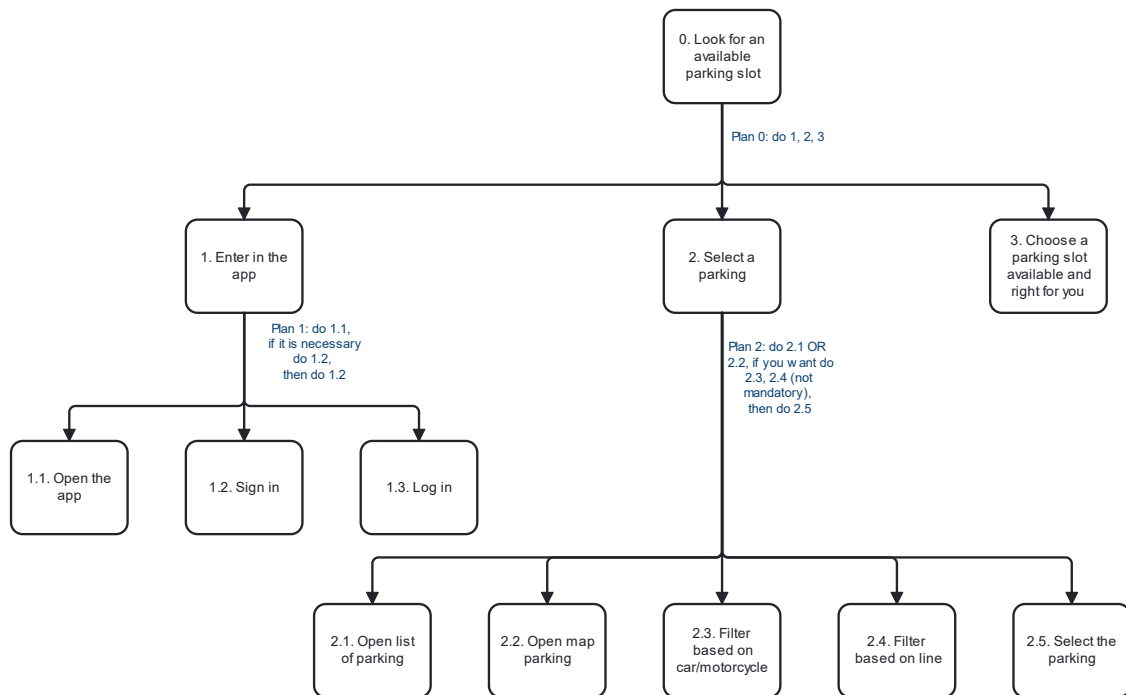
Chapter 2

Task and Dialogue analysis: HTA & STN

2.1 Hierarchical Task Analysis

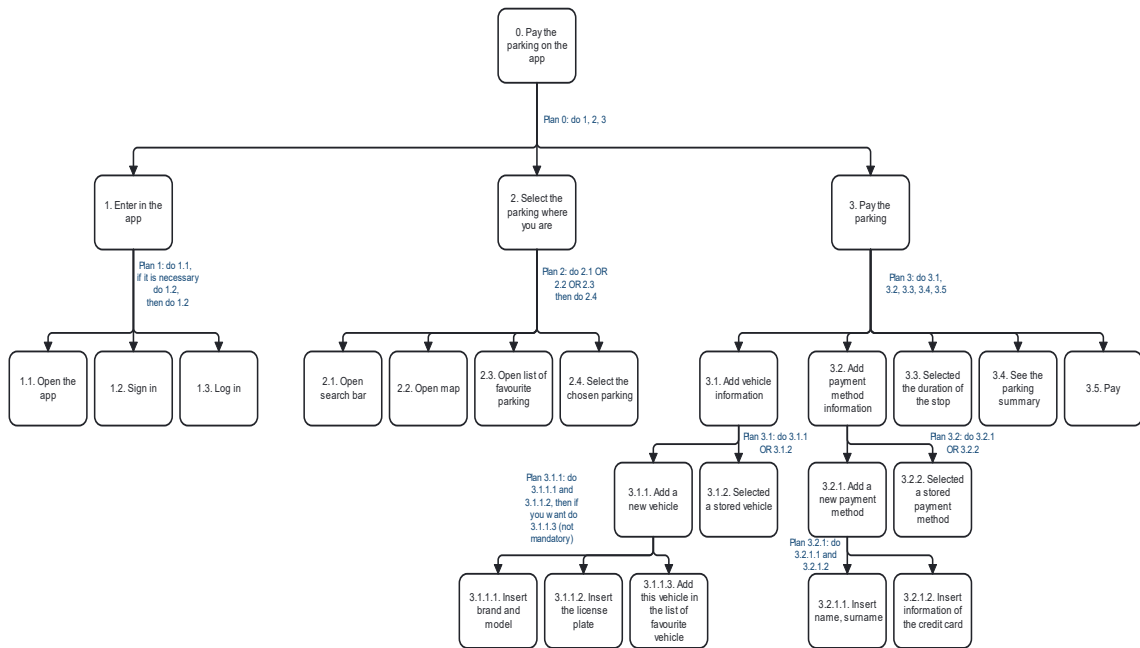
2.1.1 Look for an available parking slot

In this task, the user logs in to the app or registers if they are new. Then he must select the car park based on his preferences and based on filters, and check whether that car park has spaces available and finally choose the right slot for him, based on his needs.



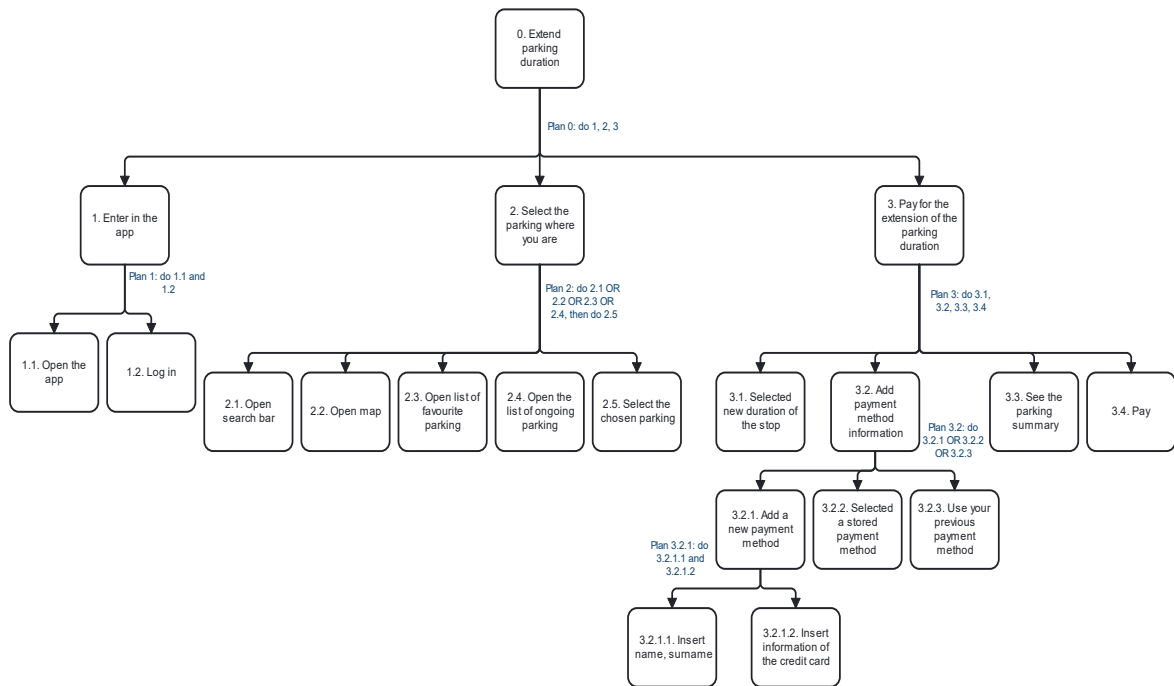
2.1.2 Pay the parking on the app

Once parked, the user pays for parking using the application. He accesses the app, selects the car park where it is located and pays for it, entering information on the duration, the vehicle and the payment method.



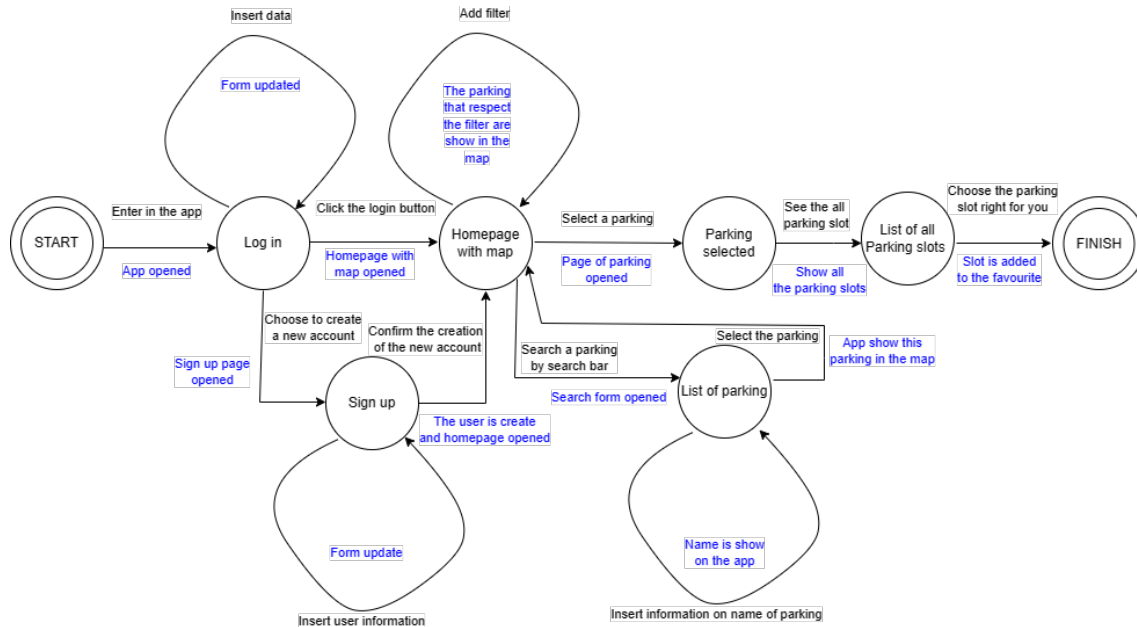
2.1.3 Extend parking duration

In the event that the user needs to increase the duration of his parking, before the current one expires and avoid the fine, he can extend the duration through the app without returning to the vehicle. To do this, enter the app, select the parking in progress through various selection alternatives and extend the parking duration, having selected the new duration and payment method.

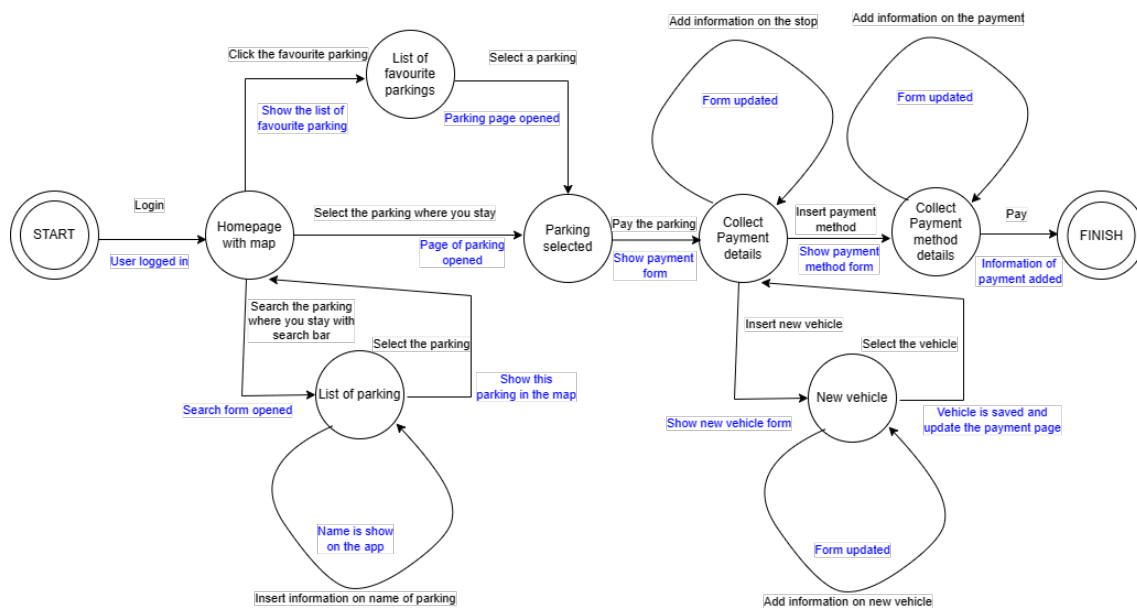


2.2 State Transition Networks

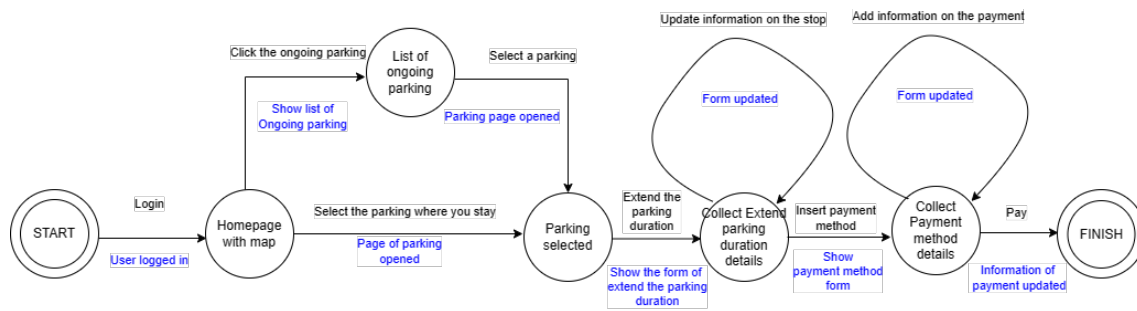
2.2.1 Look for an available parking slot



2.2.2 Pay the parking on the app



2.2.3 Extend parking duration



Chapter 3

Mockup: prototype 1

The Mockup of the first prototype was created using Balsamiq Wireframes. In this first prototype the main features of the application were represented:

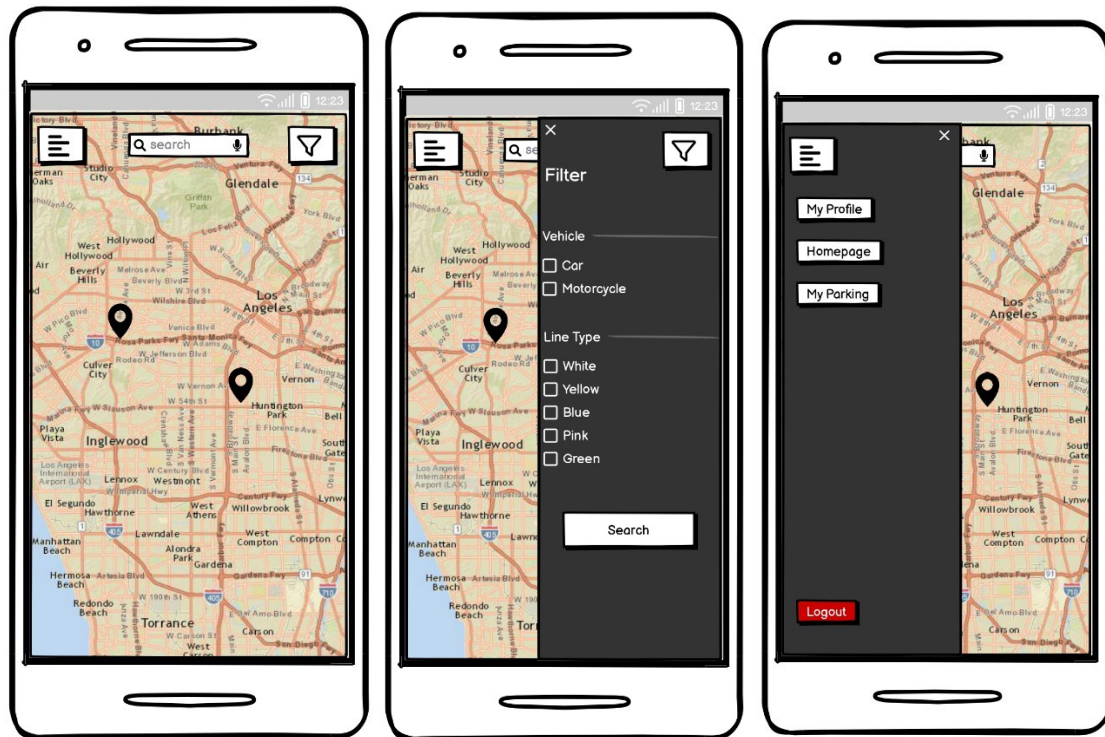
- Sign up for new users
- Login
- Search for a parking using specific filters
- Search for an available parking slot to reduce/optimize search times, possibly select a specific slot to receive info about the availability status of the slot (but not booking).
- Pay for parking directly from the app
- Extend the duration of your parking directly from the app
- Manage parking: completed, slot selected, ongoing and favorites
- Manage your profile



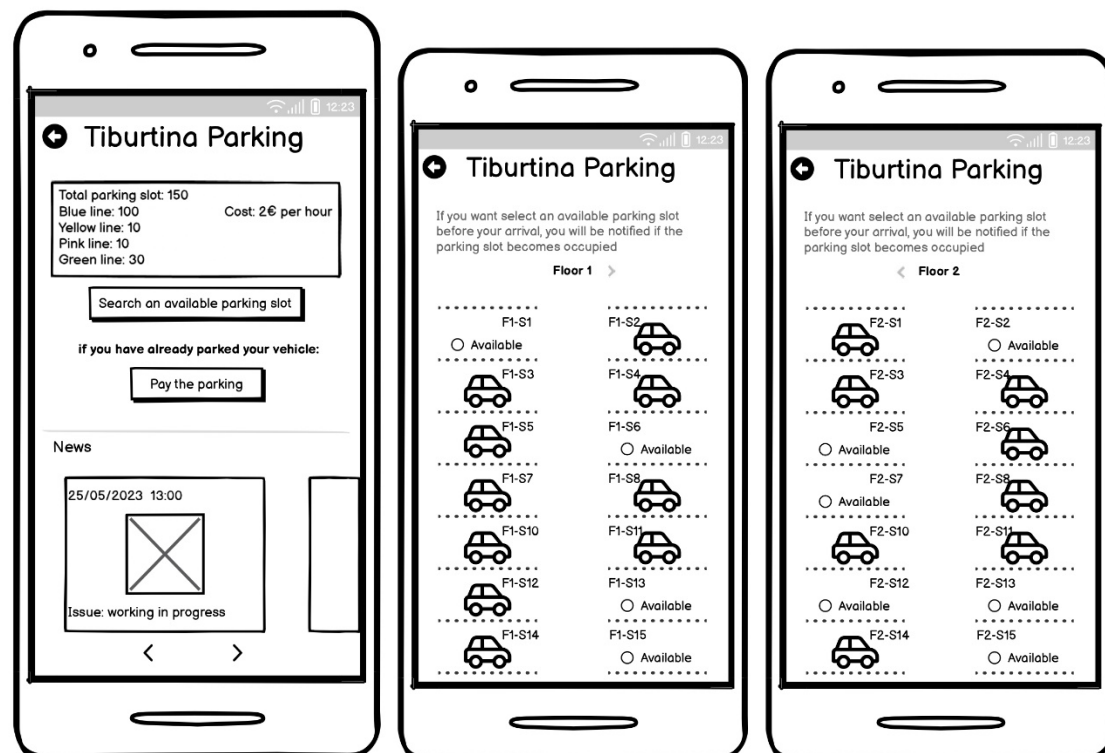
Sign up



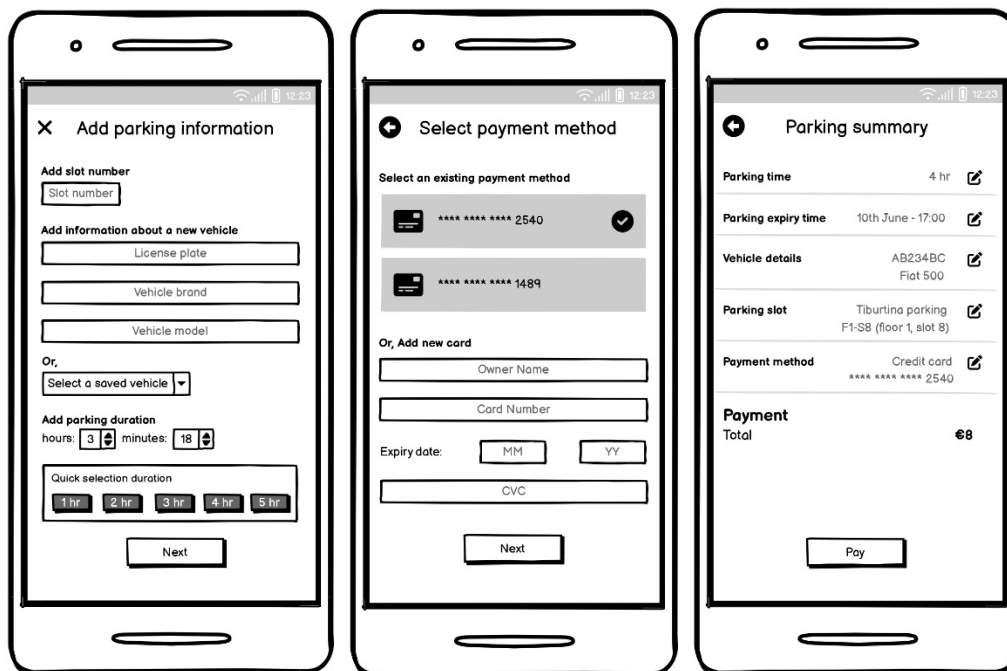
Login and my profile page



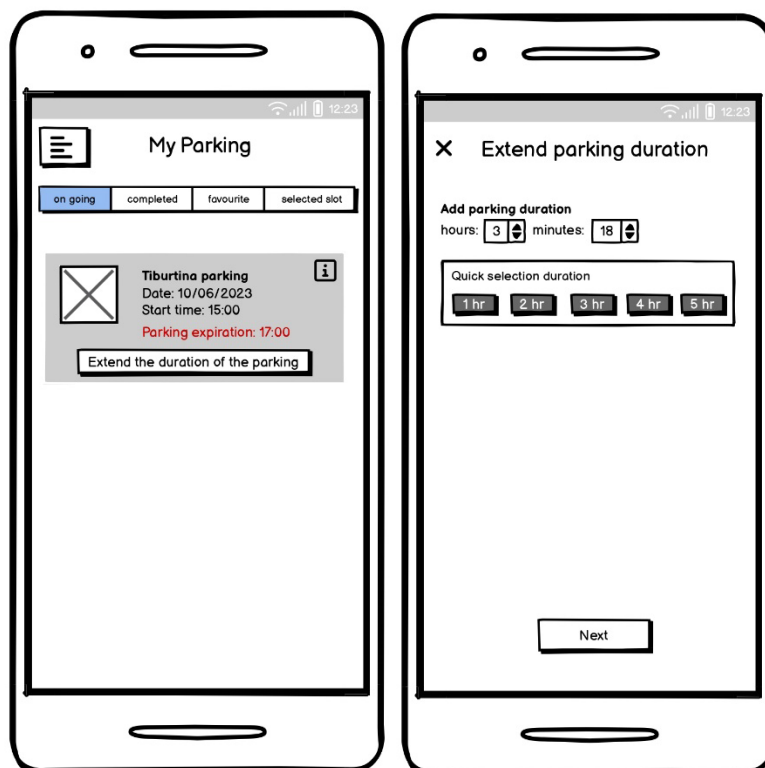
Homepage (map), filter section for parking and menù section



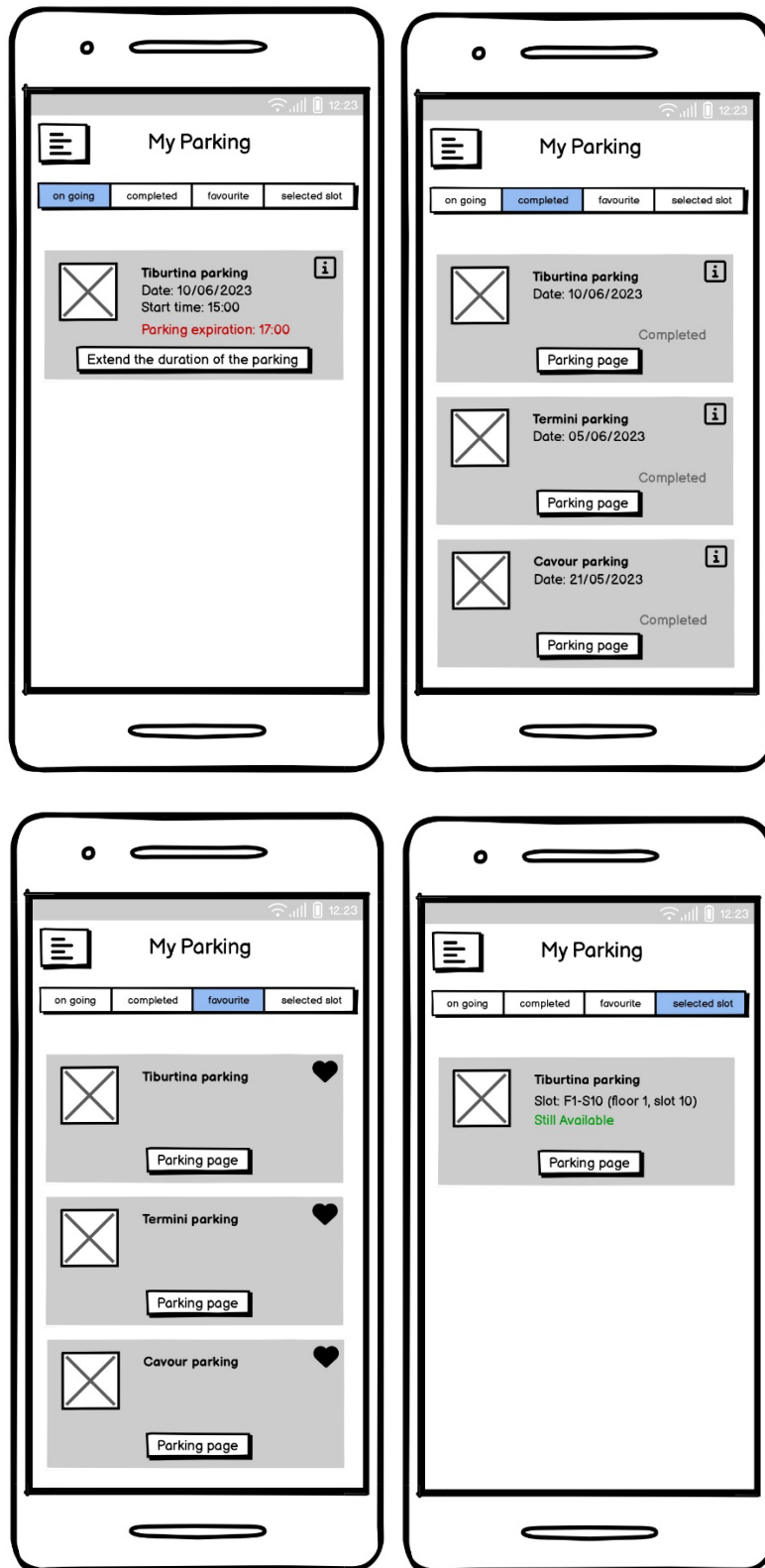
Parking page and map of the parking with all slot



Payment task with 3 page to insert all info about the payment



Extend parking duration task



My parking page with all info about ongoing, completed and favourite parking, and selected slot to receive info about the availability

Chapter 4

Expert Based Evaluation

4.1 Heuristic Evaluation

A heuristic evaluation is a method for identifying design problems in a user interface. Evaluators judge the design against a set of guidelines (called heuristics) that make systems easy to use.

A heuristic evaluation can be conducted with any set of heuristics. To assess usability, Jakob Nielsen's 10 Usability Heuristics were used, a set of high-level guidelines based on an understanding of human behavior, psychology, and information processing.

The Jakob Nielsen's 10 Usability Heuristics are:

1. **Visibility of system status:** The design should always keep users informed about what is going on, through appropriate feedback within a reasonable amount of time.
2. **Match between system and the real world:** The design should speak the users' language. Use words, phrases, and concepts familiar to the user, rather than internal jargon. Follow real-world conventions, making information appear in a natural and logical order.
3. **User control and freedom:** Users often perform actions by mistake. They need a clearly marked "emergency exit" to leave the unwanted action without having to go through an extended process.
4. **Consistency and standards:** Users should not have to wonder whether different words, situations, or actions mean the same thing. Follow platform and industry conventions.
5. **Error prevention:** Good error messages are important, but the best designs carefully prevent problems from occurring in the first place. Either eliminate error-prone conditions, or check for them and present users with a confirmation option before they commit to the action.
6. **Recognition rather than recall:** Minimize the user's memory load by making elements, actions, and options visible. The user should not have to remember information from one part of the interface to another. Information required to use the design (e.g. field labels or menu items) should be visible or easily retrievable when needed.
7. **Flexibility and efficiency of use:** Shortcuts — hidden from novice users — may speed up the interaction for the expert user so that the design can cater to both inexperienced and experienced users. Allow users to tailor frequent actions.
8. **Aesthetic and minimalist design:** Interfaces should not contain information that is irrelevant or rarely needed. Every extra unit of information in an interface

competes with the relevant units of information and diminishes their relative visibility.

9. **Help users recognize, diagnose, and recover from errors:** Error messages should be expressed in plain language (no error codes), precisely indicate the problem, and constructively suggest a solution.
10. **Help and documentation:** It's best if the system doesn't need any additional explanation. However, it may be necessary to provide documentation to help users understand how to complete their tasks.

Heuristic evaluations are useful for identifying glaring problems in an interface.

The evaluation of the first prototype was made by the Alba Bisante, who carried out the heuristic evaluation producing the following report:

Page	Heuristic violated	Severity (1-5)	Description/comment
Login	Help users recognize, diagnose, and recover from errors	4	Include a "Forgot password" link
Tiburtina Parking page	Match between system and real world, Help and documentation	4	It is not clear the description of the parking area - is it a public parking area on the street, or a private facility as the mall example in the introduction? Also, it is not clear what the line classification stands for. I assume it refers to the classification of street parking spots in Rome, but it may not be shared between all municipalities (hence between all users).

			In this case, it would suggest the parking area is public. On the other hand, the news section seems more a private thing. Having said so, the parking type is not reported in the parking slot detail, so what is its purpose?
Tiburtina Parking page	Visibility of system status, Flexibility and efficiency of use	4	Why ask the user to “search for a parking spot”, if the system already knows the situation? Which is the label if no parking spot is available?
Homepage (map)	Match between system and real world, Consistency and standards	3	Indicating something with a pin on the map usually indicates a single activity (e.g. a single and private parking facility). If it is public I would suggest indicating it with an area rather than the single pin.
Homepage	Visibility of system status	4	I would have expected a more evident feedback about having paid a parking spot. It is displayed in the dedicated section (Menu>My Parking>On going), but also in the homepage could be useful.

General comment 1	Match between the system and the real world	4	It may seem that you are allowing to book and pay for specific parking spots (e.g. the chance of selecting a parking spot in the map, but also in the payment form, there is the section for the “parking slot” selected). However, reading more carefully (e.g. the note “...you will be notified if the parking slot becomes occupied”) you understand that the app is just providing info about the availability status of the spots. I would suggest an overall clarification about this, as it is the main feature of the app.
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- Jakob Nielsen’s usability heuristics have been used.
- Severity:
 - 0 = I don’t agree that this is a usability problem at all
 - 1 = Cosmetic problem only
 - 2 = Minor usability problem
 - 3 = Major usability problem
 - 4 = Usability catastrophe

4.2 Cognitive Walkthrough

The cognitive walkthrough method is a usability inspection method used to identify usability issues in interactive systems, focusing on how easy it is for new users to accomplish tasks with the system.

For each task walkthrough considers:

- what impact will interaction have on user?
- what cognitive processes are required?
- what learning problems may occur?

Analysis focuses on goals and knowledge: does the design lead the user to generate the correct goals?

4.2.1 Tasks analyzed

The task analyzed for the CW is the payment of a parking spot. In this task there are several actions that can be performed based on the case analysed, making some actions optional; for example, when adding the vehicle there are two possibilities: add a new vehicle or select a vehicle already saved in the system (the same applies to a credit card). The other action that may vary between cases is the selection of the parking duration, which can be entered manually or select a fixed duration (quick selection duration). Therefore, in order to analyze all the possible actions for the task in question, it was decided to analyze 2 cases, emphasizing the common actions and the different actions. In particular, the common actions between the two cases are access to subsequent pages and final payment.

It is assumed that for both tasks analysed, the user has already parked in a chosen car park, and has already logged into the app, in particular in the payment section of the car park where he/she parks.

1. The first task analyzed is:

“Pay for a parking with a duration of 3:30 hours, selecting an already saved vehicle, and using your saved credit card.”

In particular, in this case, a vehicle and a credit card already saved in the system are selected, and the user enters the duration manually.

The screen displays a form titled "Add parking information" with a close button (X) in the top left. Below the title, there is a section "Add information about a new vehicle" with three input fields: "License plate", "Vehicle brand", and "Vehicle model". Below these fields is the text "Or," followed by a dropdown menu labeled "Select a saved vehicle". Underneath is a section "Add parking duration" with "hours:" and "minutes:" labels, each followed by a numeric input field (3 and 18 respectively). Below this is a "Quick selection duration" section with five buttons: "1 hr", "2 hr", "3 hr", "4 hr", and "5 hr". At the bottom of the form is a "Next" button.

Action 1: Select an already saved vehicle

Response 1: The system shows the selected vehicle.

Action 2: Select the duration of the parking on 3:30 hours.

Response 2: The system shows the selected time duration of the parking.

Action 3: Press the “Next” button to access the next page.

Response 3: The system shows the next page to select the payment method.

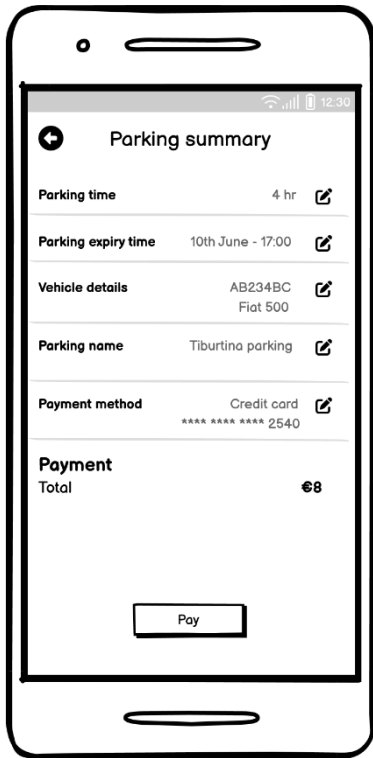
The screen displays a form titled "Select payment method" with a back button in the top left. Below the title is a section "Select an existing payment method" with two card options, each showing a card icon, a masked number (**** * 2540 and **** * 1489), and a checkmark. Below this is the text "Or, Add new card" followed by four input fields: "Owner Name", "Card Number", "Expiry date:" (with "MM" and "YY" sub-fields), and "CVC". At the bottom of the form is a "Next" button.

Action 4: Select your saved credit card.

Response 4: The system shows the selected credit card.

Action 5: Press the “Next” button to access the next page.

Response 5: The system shows the next page with a parking summary.



Action 6: Press the “Pay” button to pay the parking.

Response 6: The data is collected by the system and the parking is paid correctly.

2. The second task analyzed is:

“Pay for a parking with a duration of 2:00 hours, using a new vehicle: 'Volkswagen' car, model 'T Roc', license plate 'FB619ZY', and using a new credit card to pay.”

In particular, in this case, a new vehicle and a new credit card are inserted, and the user uses the “quick selection duration” section to set the parking duration.

Action 1: Press text field input and digit the license plate FB619ZY

Response 1: Each digit is displayed as typed, flashing cursor moves to next position.

Action 2: Press text field input and digit the vehicle brand Volkswagen.

Response 2: Each digit is displayed as typed, flashing cursor moves to next position.

Action 3: Press text field input and digit the vehicle model TRoc.

Response 3: Each digit is displayed as typed, flashing cursor moves to next position.

Action 4: Select the duration of the parking on 2:00 hours, using “quick selection duration”.

Response 4: The system shows the selected time duration of the parking.

Action 5: Press the “Next” button to acces the next page.

Action 6: Press text field input and digit the Owner Name “Lorenzo Russo”.

Response 6: Each digit is displayed as typed, flashing cursor moves to next position.

Action 7: Press text field input and digit the Card Number “12345678”.

Response 7: Each digit is displayed as typed, flashing cursor moves to next position.

Action 8: Press text field input and digit the Expiry date (month and year), “10/2028”.

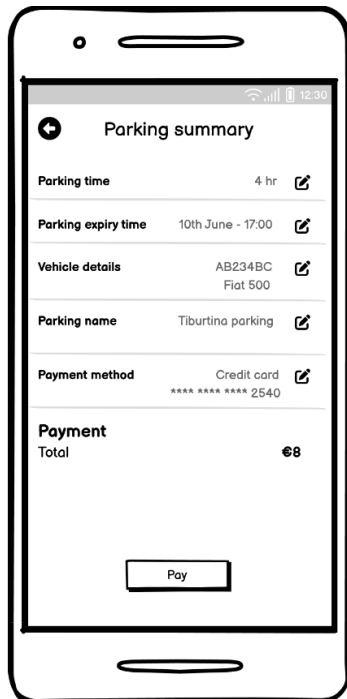
Response 8: Each digit is displayed as typed, flashing cursor moves to next position.

Action 9: Press text field input and digit the CVC code “123”.

Response 9: Each digit is displayed as typed, flashing cursor moves to next position.

Action 10: Press the “Next” button to acces the next page.

Response 10: The system shows the next page with a parking summary.



Action 11: Press the “Pay” button to pay the parking.

Response 11: The data is collected by the system and the parking is paid correctly.

5.2.2 Expert Report

The evaluator of the Cognitive Walkthrough is Valeria Mirabella.

For each action the expert answered the following questions:

1. Is the effect of the action the same as the user’s goal at that point?
2. Will users see that the action is available?
3. Once users have found the correct action, will they know it is the one they need?
4. After the action is taken, will users understand the feedback they get?

1. The first task:

“Pay for a parking with a duration of 3:30 hours, selecting an already saved vehicle, and using your saved credit card.”

Action 1: Select an already saved vehicle

Response 1: The system shows the selected vehicle.

A1: yes

A2: yes

A3: yes

A4: looking at the material it is not possible to individuate the system reaction, but it is reasonable that the vehicle will be selected from a drop-down list and will be visible in the interface.

Action 2: Select the duration of the parking on 3:30 hours.

Response 2: The system shows the selected time duration of the parking.

A1: yes

A2: yes

A3: yes

A4: Yes

Action 3: Press the “Next” button to access the next page.

Response 3: The system shows the next page to select the payment method.

A1: yes

A2: yes

A3: yes

A4: Yes

Action 4: Select your saved credit card.

Response 4: The system shows the selected credit card.

A1: yes

A2: yes

A3: yes

A4: Yes

Action 5: Press the “Next” button to access the next page.

Response 5: The system shows the next page with a parking summary.

A1: yes

A2: yes

A3: yes

A4: Yes

Action 6: Press the “Pay” button to pay the parking.

Response 6: The data is collected by the system and the parking is paid correctly.

A1: yes

A2: yes

A3: yes

A4: looking at the material it is not possible to individuate the system feedback.

2. The second task analyzed is:

“Pay for a parking with a duration of 2:00 hours, using a new vehicle: 'Volkswagen' car, model 'T Roc', license plate 'FB619ZY', and using a new credit card to pay.”

Action 1: Press text field input and digit the license plate FB619ZY

Response 1: Each digit is displayed as typed, flashing cursor moves to next position.

A1: yes

A2: yes

A3: yes

A4: Yes

Action 2: Press text field input and digit the vehicle brand Volkswagen.

Response 2: Each digit is displayed as typed, flashing cursor moves to next position.

A1: yes

A2: yes

A3: yes

A4: Yes

Action 3: Press text field input and digit the vehicle model TRoc.

Response 3: Each digit is displayed as typed, flashing cursor moves to next position.

A1: yes

A2: yes

A3: yes

A4: Yes

Action 4: Select the duration of the parking on 2:00 hours, using “quick selection duration”.

Response 4: The system shows the selected time duration of the parking.

A1: yes

A2: yes

A3: yes

A4: Yes

Action 5: Press the “Next” button to access the next page.

Response 5: The system shows the next page to select the payment method.

A1: yes

A2: yes

A3: yes

A4: Yes

Action 6: Press text field input and digit the Owner Name “Lorenzo Russo”.

Response 6: Each digit is displayed as typed, flashing cursor moves to next position.

A1: yes

A2: yes

A3: yes

A4: Yes

Action 7: Press text field input and digit the Card Number “12345678”.

Response 7: Each digit is displayed as typed, flashing cursor moves to next position.

A1: yes

A2: yes

A3: yes

A4: Yes

Action 8: Press text field input and digit the Expiry date (month and year), “10/2028”.

Response 8: Each digit is displayed as typed, flashing cursor moves to next position.

A1: yes

A2: yes

A3: yes

A4: Yes

Action 9: Press text field input and digit the CVC code “123”.

Response 9: Each digit is displayed as typed, flashing cursor moves to next position.

A1: yes

A2: yes

A3: yes

A4: Yes

Action 10: Press the “Next” button to access the next page.

Response 10: The system shows the next page with a parking

A1: yes

A2: yes

A3: yes

A4: Yes

Action 11: Press the “Pay” button to pay the parking.

Response 11: The data is collected by the system and the parking is paid correctly.

A1: yes

A2: yes

A3: yes

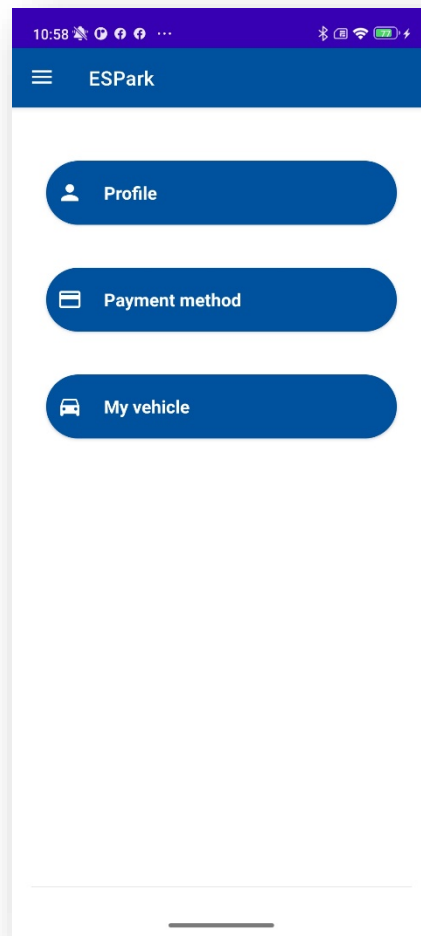
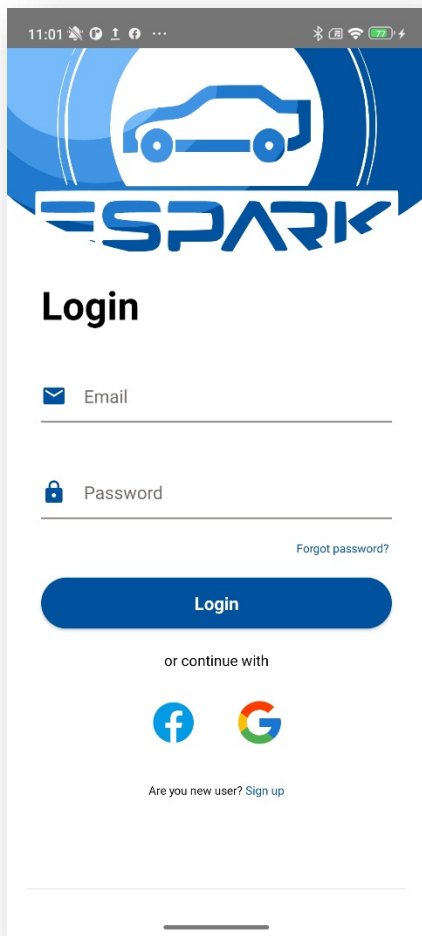
A4: looking at the material it is not possible to individuate the system feedback.

4.3 Prototype 2

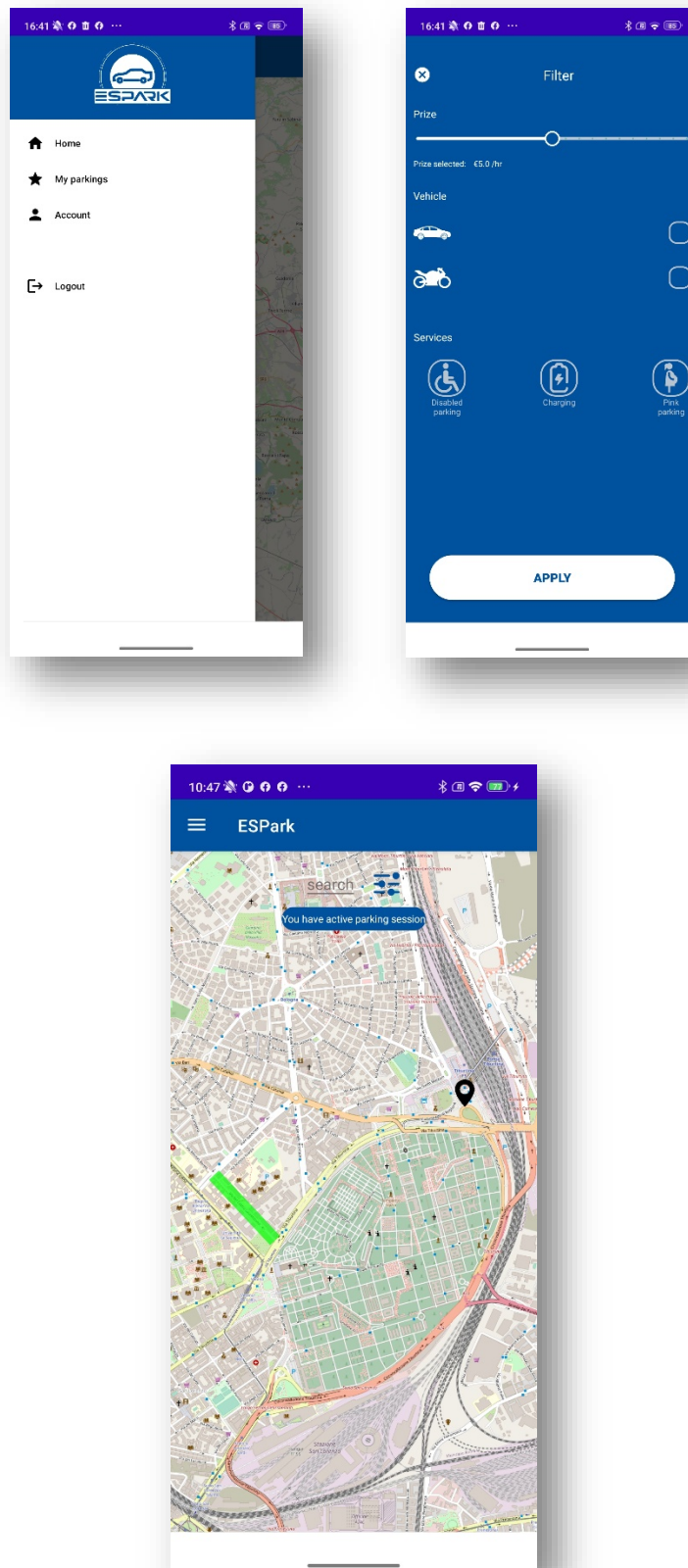
This second prototype was created with Android Studio, modifying prototype 1. This prototype was created by correcting the problems found in the Expert Based Evaluation (HE and CW) phase.

The image displays two mobile application prototypes side-by-side. The left prototype, titled 'Create your Account', features a purple status bar at the top with the time 16:39. It includes a close button (X), a title, and input fields for 'Email' and 'Password'. A blue 'Sign up' button is positioned below the inputs. Further down, it offers social login options with Facebook and Google icons, preceded by the text 'or continue with'. At the bottom, a link reads 'Already have an account? Sign in'. The right prototype, titled 'Fill Your Profile', has a purple status bar showing 16:40. It features a back arrow and a title. Below are input fields for 'Name', 'Surname', 'Date of Birth', and 'Phone Number'. A blue 'SIGN UP' button is at the bottom.

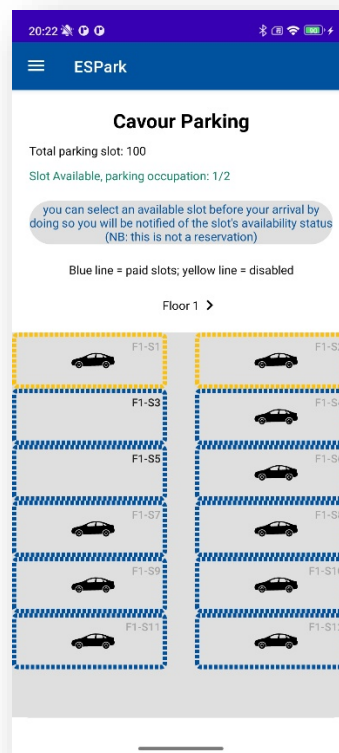
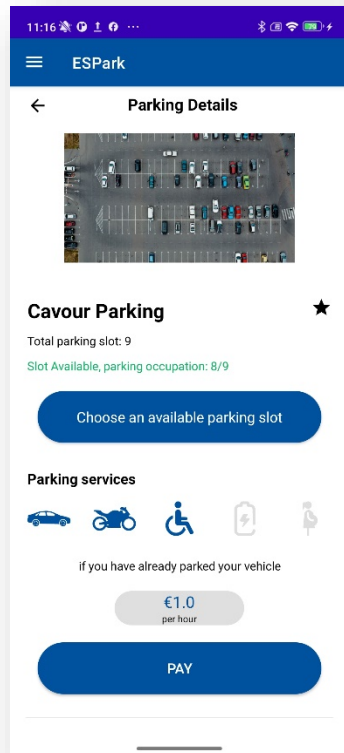
Sign up



Login and my profile page



Homepage (map), filter section for parking and menù section



Parking page and map of the parking with all slot

10:57

...



Add payment information

Choose your payment method

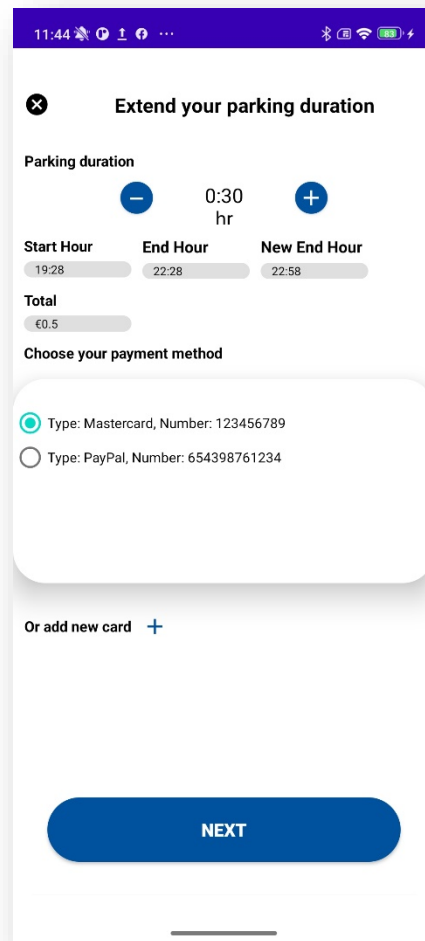
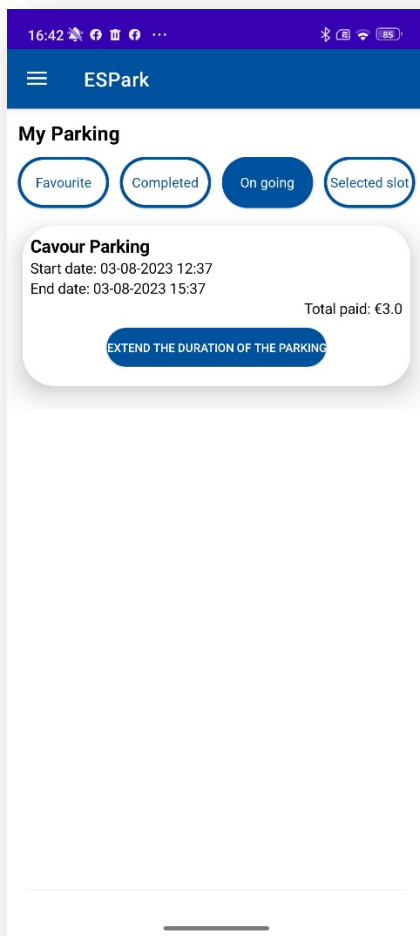
☒ Type: Mastercard, Number: 123456789

☐ Type: PayPal, Number: 654398761234

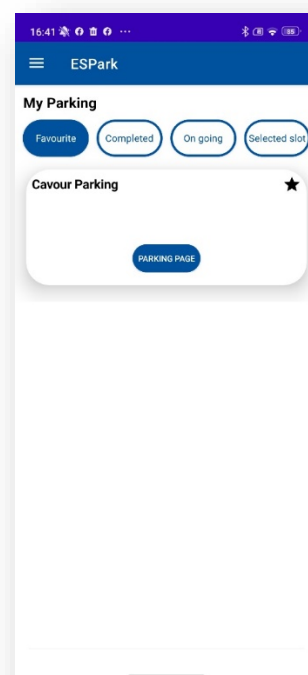
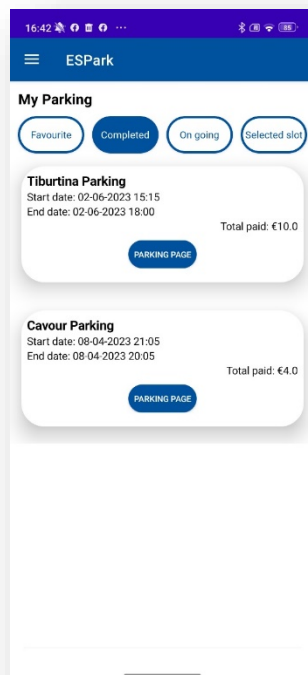
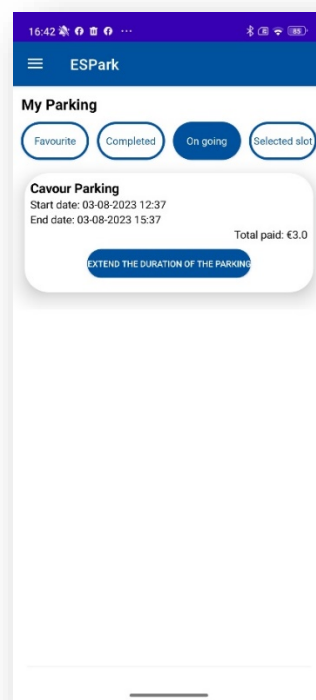
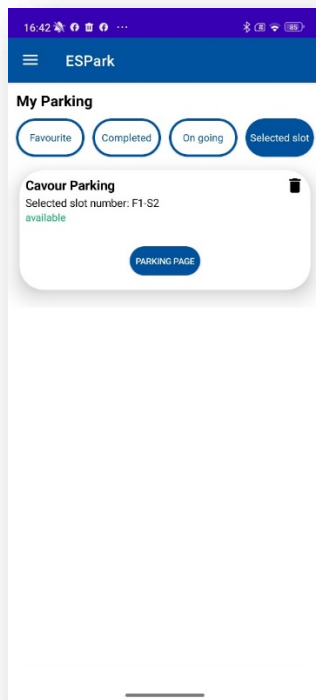
Or add new card 

NEXT

Payment task with 3 page to insert all info about the payment



Extend parking duration task



My parking page with all info about ongoing, completed and favourite parking, and selected slot to receive info about the availability

Chapter 5

User Based Evaluation

5.1 Think Aloud

"Think-aloud" is a cognitive technique used in usability testing and cognitive psychology. It involves asking participants to verbalize their thoughts, feelings, and decision-making processes as they perform a task or interact with a system. Think-aloud is a valuable tool for gaining insights into a user's thought processes and understanding how they navigate through a task or interface.

The test was carried out with 10 participants who all fell within the user profile of this application.

Participants were asked to individually perform the 3 main tasks of this application:

- Look for a parking using the filters and then look for an available parking slot in the chosen parking
- Pay the parking
- Extend parking duration

All the people we have chosen have shown great enthusiasm for the interface which is considered very simple and easy to use. Most of the participants were very enthusiastic in knowing that ideally in every parking lot there should be sensors that say with certainty whether the parking space is available or not, considering the project a great opportunity for the harmful problem of finding a parking space in big cities. Those who were less familiar with mobile applications were more skeptical about looking for parking with an application, but nevertheless they found the application very interesting, with a simple and intuitive interface with clear buttons.

The task that has created the most problems is the search for a free parking space, many users have had difficulty understanding the task. For this reason, a solution has been found to ensure that users are able to perform and understand the task in an optimal way; an "Info" button has been inserted near the "Look for an available parking slot" button so that the user can press the button to obtain more detailed information on the service offered. Another problem has been noticed by users and is the following: after selecting the desired slot there is nothing to tell us what was done by clicking it; so we decided to introduce a message after selecting the slot that explains everything

Another problem encountered by users is the vision of the car parks after using the filters; the solution adopted is to display on the map only the car parks that comply with the entered filters.

The last suggestion from users was to enter the street where the car park is located.

There were no problems with paying for parking. Finally, to extend the duration of the parking, no user found problems completing the task.

5.2 Controlled Experiment

A controlled experiment is an experiment in which an observer tests a hypothesis by looking for changes brought on by alterations to certain variables. In a controlled experiment, an independent variable (e.g. different interface styles) is the only factor that is allowed to be manipulated, with the dependent variable (e.g. time to complete a task) as the factor that the independent variable will affect. The main advantage of a controlled experiment is that it is easier to eliminate uncertainty about the significance of the results.

In the experiment, we would like to prove that our prediction is correct i.e. to confirm our hypothesis. This is done by disproving/rejecting the null hypothesis (a null hypothesis states that a variation in the Independent Variables does not make any differences in the Dependent Variables). Moreover, to disprove this null hypothesis and to demonstrate that our hypothesis is true, it is necessary as first thing that the data collected during the experiment have a statistical relevance (this demonstrates that the null hypothesis is false), and then, if the first thing is true, it is necessary to observe the differences among the means of the data collected and eventually show that these differences prove our hypothesis.

5.2.1 Experiment 1

In this first experiment the following task was studied: search for a parking space with available slots using the map.

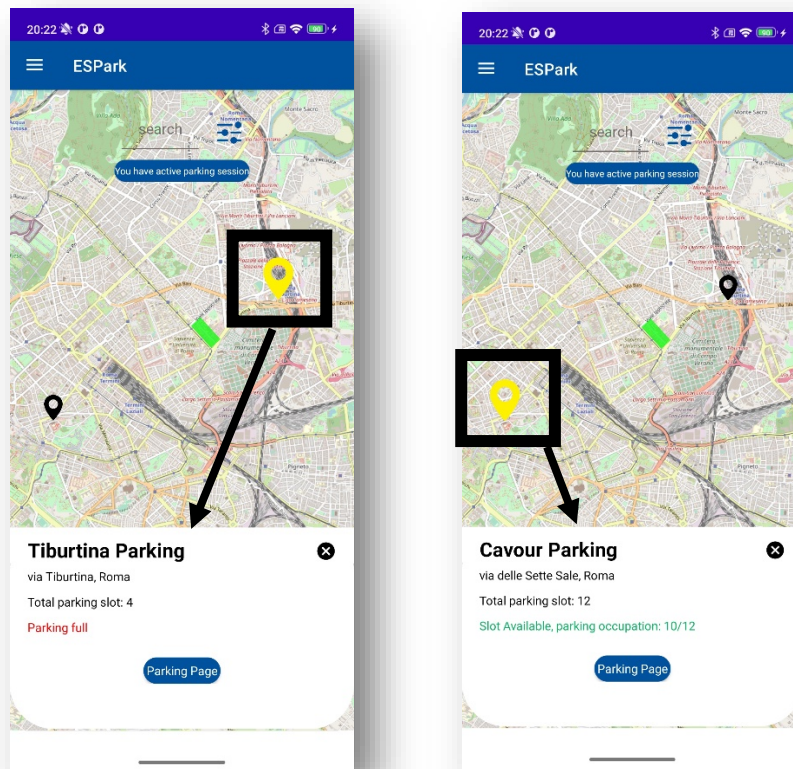
Dependent variables

The dependent variable was measured as the time (in seconds) taken by each user to complete the task.

Independent variables

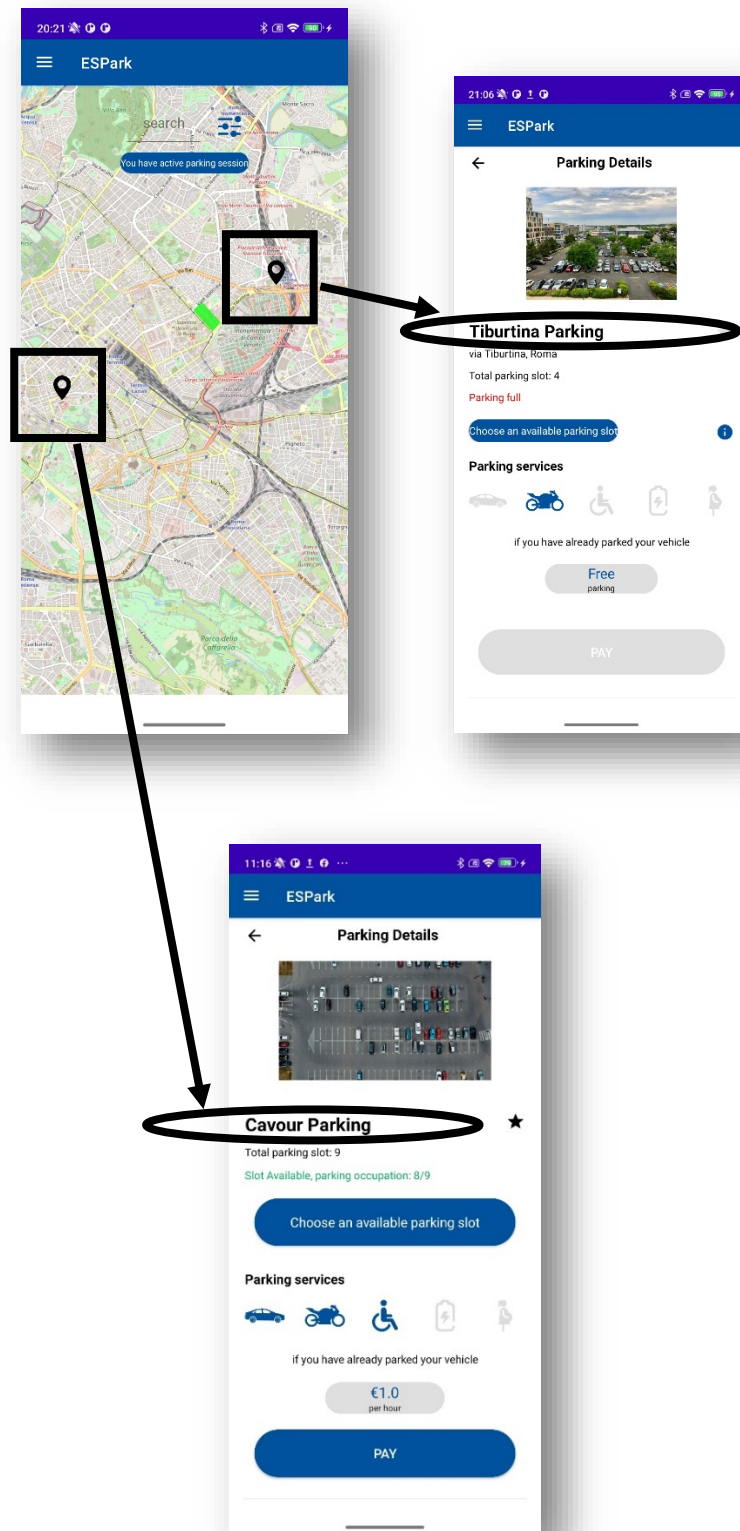
Two different interfaces were used as independent variables:

- *Interface 1*: in this interface there is a small screen preceding the page of the selected car park, where it is possible to see some information about the car park including availability information.



Interface 1

- *Interface 2*: in this interface there is no screen preceding the page of the selected car park; therefore, to see whether a slot is available or not, the user must access the parking page.



Interface 2

Hypotheses

The hypothesis is a prediction of the outcome of the experiment. The experiment aims to prove the hypothesis disproving the null hypothesis (no difference in the dependent variables caused by changes in the independent variables).

- *Null hypothesis:* both interfaces offer the same performance to users without any distinction; in particular, the time taken to execute the task in interface 1 is more or less the same using interface 2.
- *Our hypothesis:* interface 1 offers better performance than interface 2; it is more intuitive and faster to search for an available parking using interface 1 than interface 2.

Running experiment

The between-group method was used, asking 10 participants divided into two groups of 5 to perform the task; in particular, one group used interface 1 and the other group used interface 2.

In this experiment we collect all the times (in seconds) that each user takes to finish the task, using a stopwatch.

5.2.1.1 ANOVA single factor analysis and results

Once all the relevant data has been collected, we continue with the execution of the single-factor statistical analysis ANOVA (analysis of variance). Single factor ANOVA is a statistical analysis used to determine whether statistically significant differences exist between different data.

Below is the ANOVA analysis:

Interface 1	Interface 2
7,5	25,1
16,2	30,2
10,8	16,8
8,3	20,3
6,5	11,7

Analisi varianza: ad un fattore

RIEPILOGO				
Gruppi	Conteggio	Somma	Media	Varianza
Colonna 1	5	49,3	9,86	15,093
Colonna 2	5	104,1	20,82	51,477

ANALISI VARIANZA						
Origine della variazione	SQ	gdl	MQ	F	Valore di significatività	F crit
Tra gruppi	300,304	1	300,304	9,022202	0,016975743	5,317655
In gruppi	266,28	8	33,285			
Totale	566,584	9				

From the analysis we obtained $F > F_{crit}$, i.e. $9.022202 > 5.317655$. Thanks to this result obtained we can reject the null hypothesis; this means that the data we analyzed have statistical significance and that the two populations analyzed (the times taken to complete the task in interface 1 and the times taken to complete the same task in interface 2) are not equal. So from this we get that using interface 1 or interface 2 is not the same thing.

For this reason, interface 1 is present in the final version of the app.

5.2.2 Experiment 2

In this second experiment the following task was studied: extend the duration of your parking directly from the app.

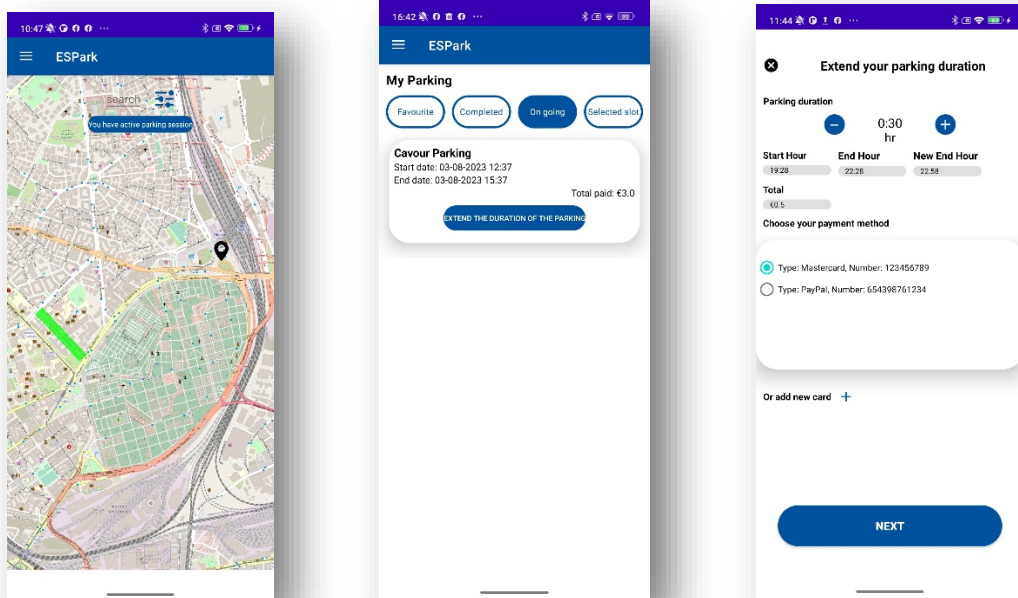
Dependent variables

The dependent variable was measured as the time (in seconds) taken by each user to complete the task.

Independent variables

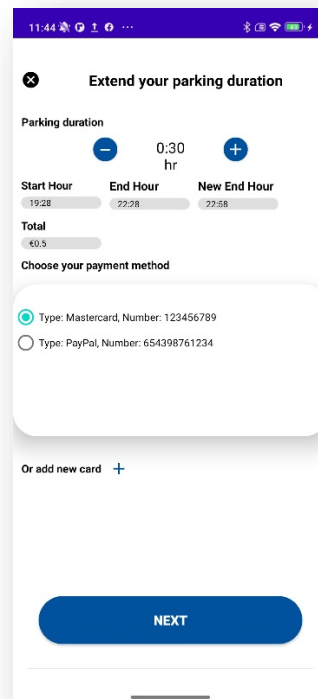
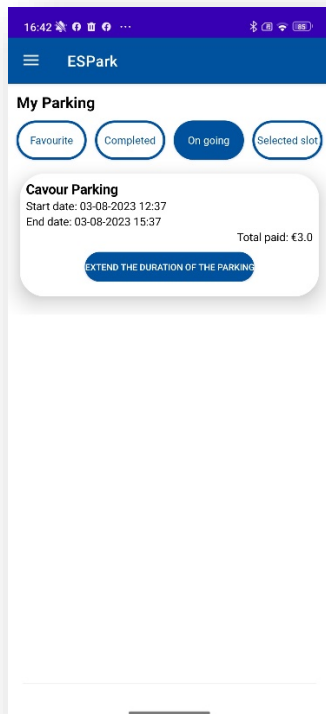
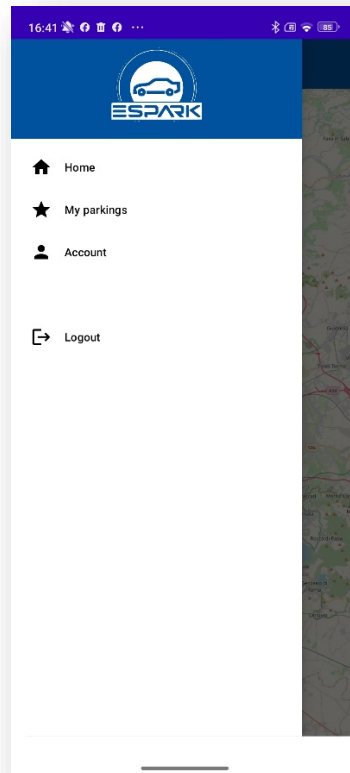
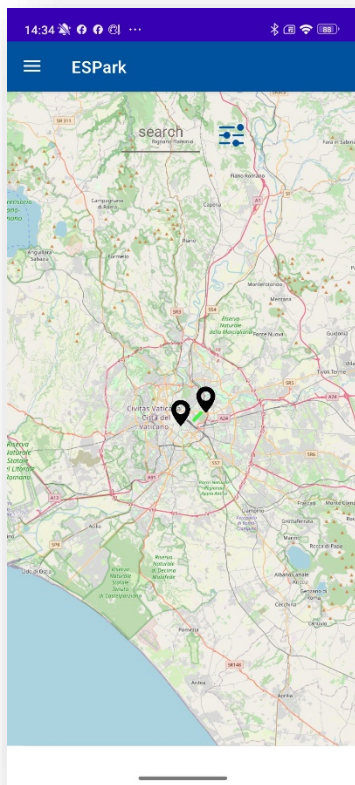
Two different interfaces were used as independent variables:

- *Interface 1*: in this interface on the homepage (page with the map view) there is a textbox that informs the user of a stop in progress, pressing it will extend the duration of the parking.



Interface 1

- *Interface 2*: in this interface instead there is no textbox in the homepage; therefore, to extend the duration of the parking, the user must access the "My parkings" page from the menu.



Interface 2

Hypotheses

The hypothesis is a prediction of the outcome of the experiment. The experiment aims to prove the hypothesis disproving the null hypothesis (no difference in the dependent variables caused by changes in the independent variables).

- *Null hypothesis:* both interfaces offer the same performance to users without any distinction; in particular, the time taken to execute the task in interface 1 is more or less the same using interface 2.
- *Our hypothesis:* interface 1 offers better performance than interface 2; it is more intuitive and faster to extend parking duration using interface 1 than interface 2.

Running experiment

The within-group method was used, asking 10 participants to complete the task in both interfaces.

Using this method is advantageous in our case because we have just 10 participants, compared to the between-group method in which a higher number of participants is needed. The disadvantage of using this method is called transfer of learning issue; that is, a user who carries out the same task twice, despite the fact that they are two different interfaces, learns additional knowledge caused by the execution of the first task which will inevitably influence the execution of the second task.

To reduce this type of problem, we chose to divide the group of participants into two: 5 participants perform the task with interface 1 and then with interface 2, the other 5 participants perform the task in the reverse order of interfaces, first in interface 1 and then in interface 2.

In this experiment we collect all the times (in seconds) that each user takes to finish the task, using a stopwatch.

5.2.2.1 ANOVA single factor analysis and results

Once all the relevant data has been collected, we continue with the execution of the single-factor statistical analysis ANOVA (analysis of variance). Single factor ANOVA is a statistical analysis used to determine whether statistically significant differences exist between different data.

Below is the ANOVA analysis:

Interface 1	Interface 2
8,7	28,7
9,5	35,6
6,5	24,2
10,7	41,3
7,8	23,4
8,6	18,5
9,1	15,7
11,8	25,8
10,3	22,9
13,9	27,2

Analisi varianza: ad un fattore

RIEPILOGO

Gruppi	Conteggio	Somma	Media	Varianza
Colonna 1	10	96,9	9,69	4,4298889
Colonna 2	10	263,3	26,33	57,320111

ANALISI VARIANZA

Origine della variazione	SQ	gdl	MQ	F	Valore di significatività	F crit
Tra gruppi	1384,448	1	1384,448	44,840421	2,79579E-06	4,413873
In gruppi	555,75	18	30,875			
Totale	1940,198	19				

From the analysis we obtained $F > F_{crit}$, i.e. $44.840421 > 4.413873$. Thanks to this result obtained we can reject the null hypothesis; this means that the data we analyzed have statistical significance and that the two populations analyzed (the times taken to complete the task in interface 1 and the times taken to complete the same task in interface 2) are not equal. So from this we get that using interface 1 or interface 2 is not the same thing.

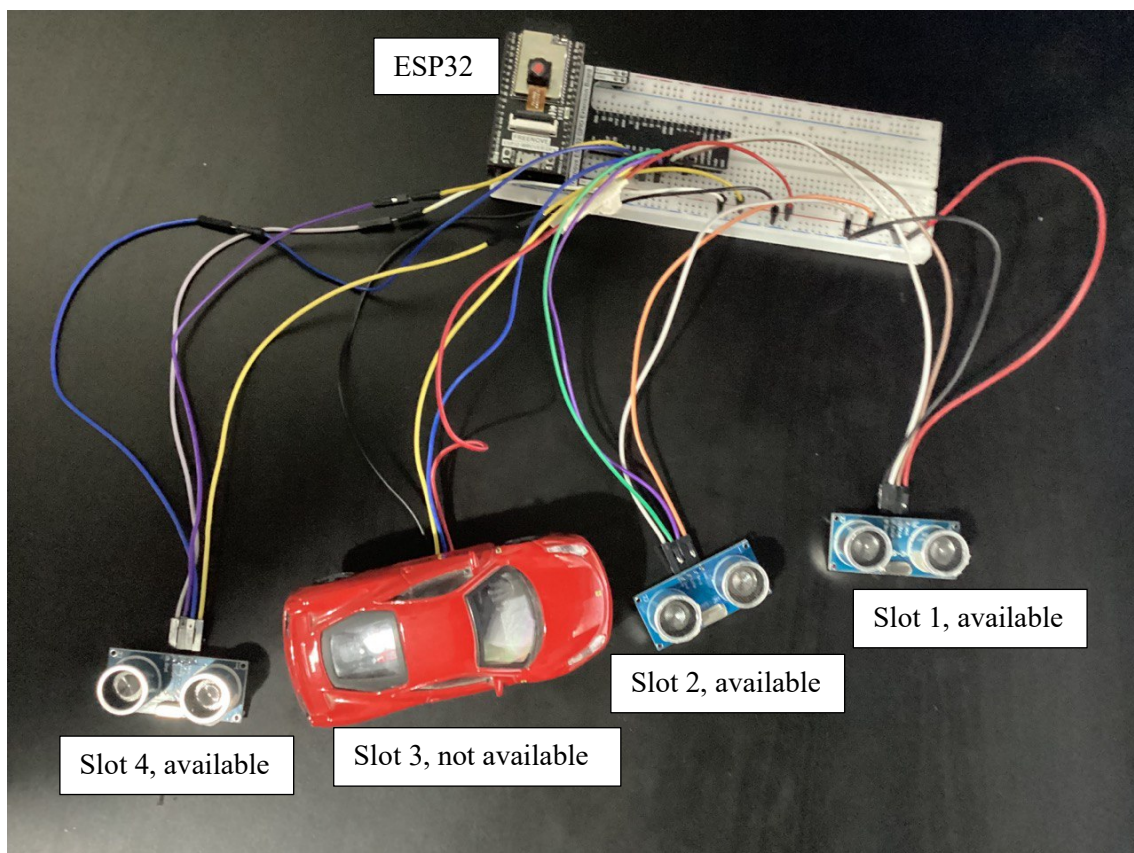
Chapter 6

Implementation and Final Product

The project was implemented by creating a working app that interfaces with the ESP32 microcontroller which simulates a real car park where there are several sensors representing the car slots, in such a way as to simulate the availability or otherwise of the slot in real time. In addition to the implementation of the app with Android Studio, a working backend was also implemented. This is a really working Android application with both front-end and back-end and with parking prototype thanks to ESP32.

6.1 Hardware implementation

To create the hardware part and therefore the simulation of real parking, the ESP32 microcontroller was used with 4 ultrasonic distance sensors to detect when a car is parked or not in that slot, and therefore communicate to all users of the app a real-time update of whether slots are available or not. The parking prototype is depicted in the figure:

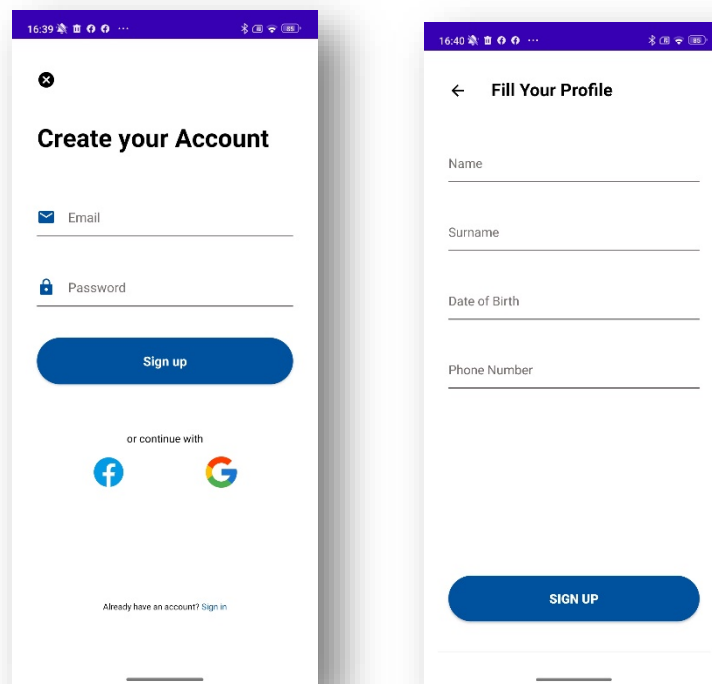


Once the prototype was created, the C code was written to program the microcontroller and sensors; to do this Arduino IDE was used.

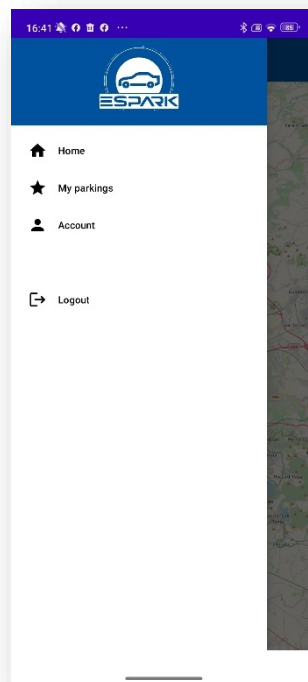
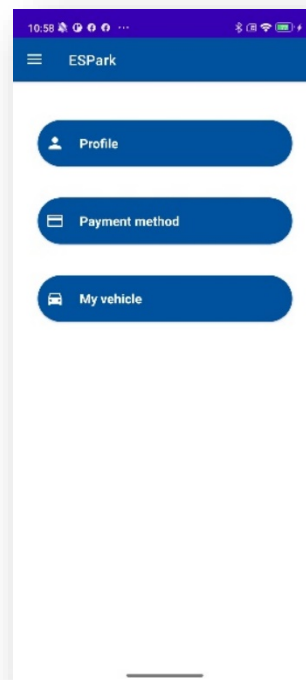
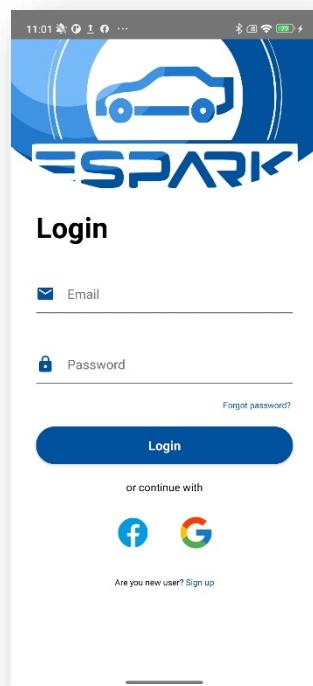
6.2 Software implementation

Through the use of Firebase, a Realtime Database and a system for authenticating users were created. Thanks to the use of a Realtime Database, in addition to storing parking and user data, it was possible to implement real-time visualization of available slots; when the microcontroller detects a change in status in a slot (from available to occupied, or viceversa) it sends this data to the Database in real-time, in this way all users of the app will be updated in real-time on the availability of parking spaces.

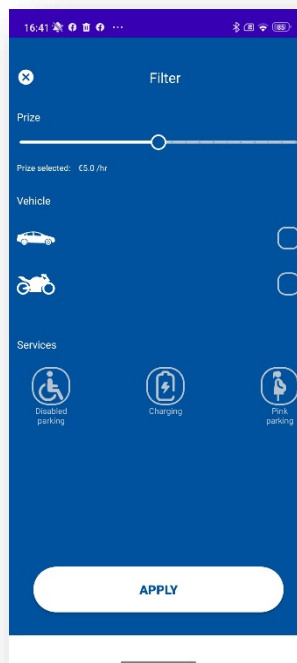
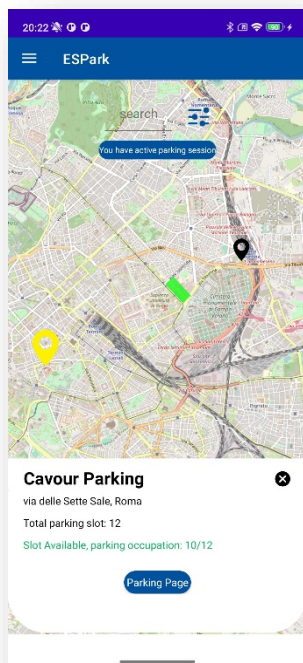
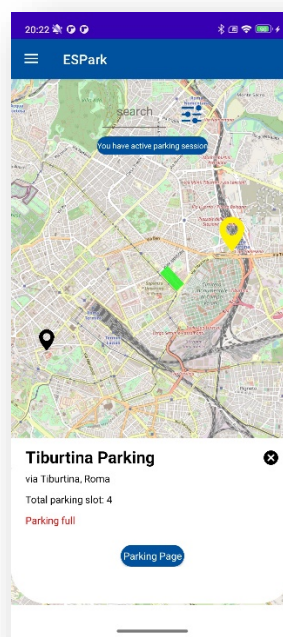
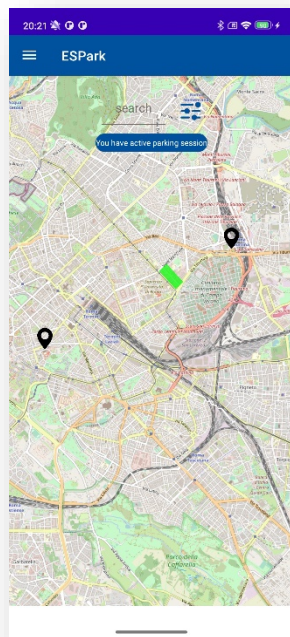
The app was created using Android Studio. Below you can see the app and all or at least the most important features created.



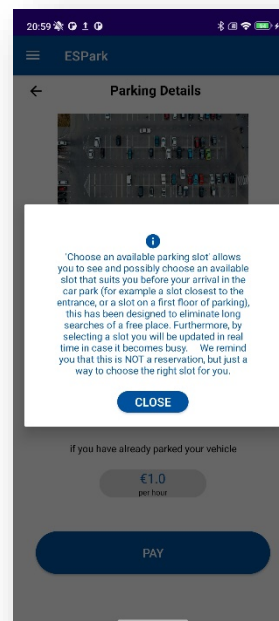
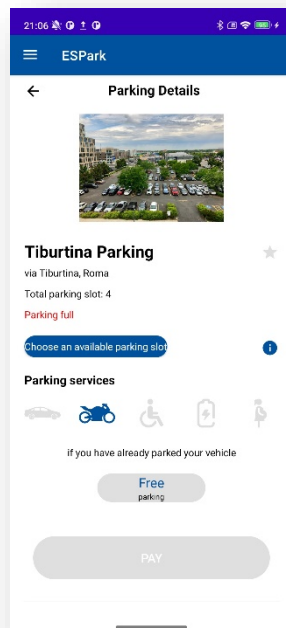
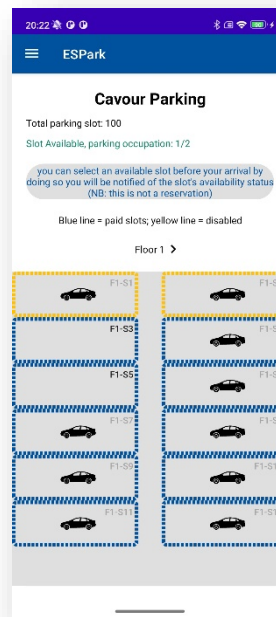
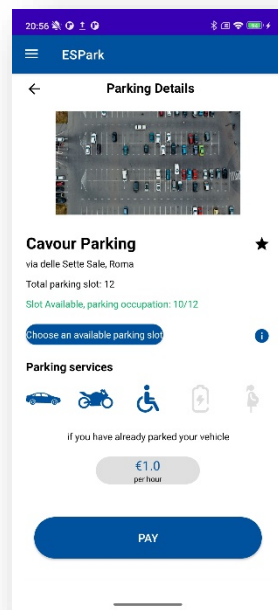
Sign up



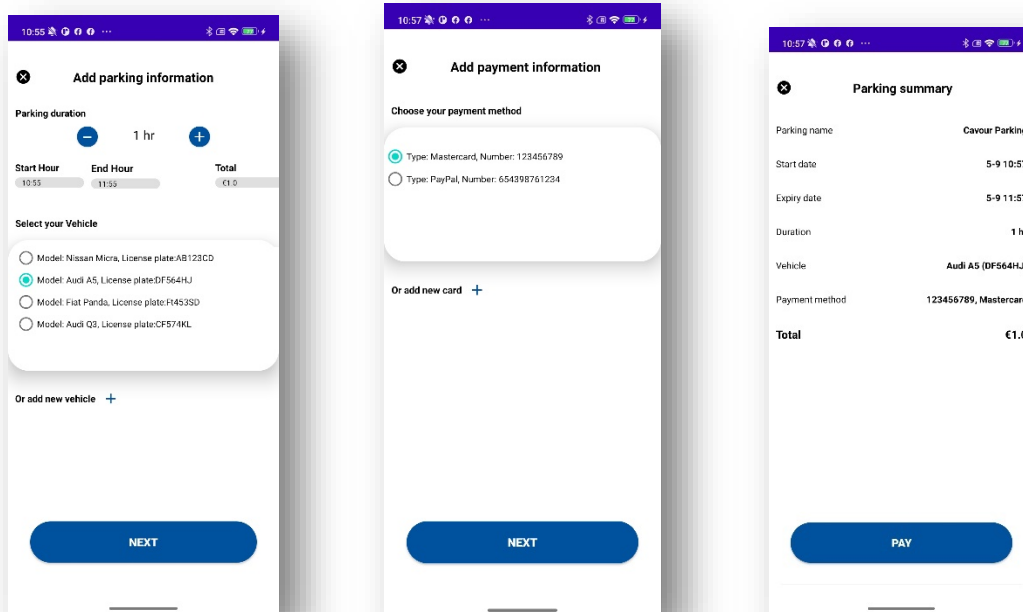
Login, my profile page and menù section



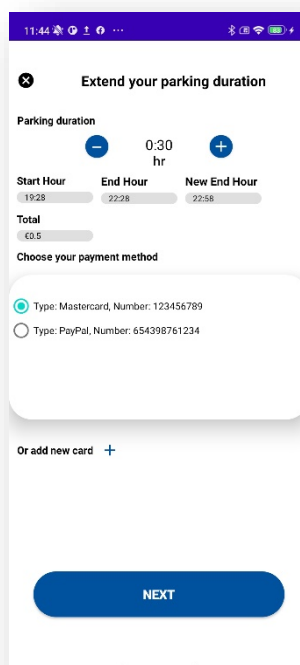
Homepage (map), filter section for parking



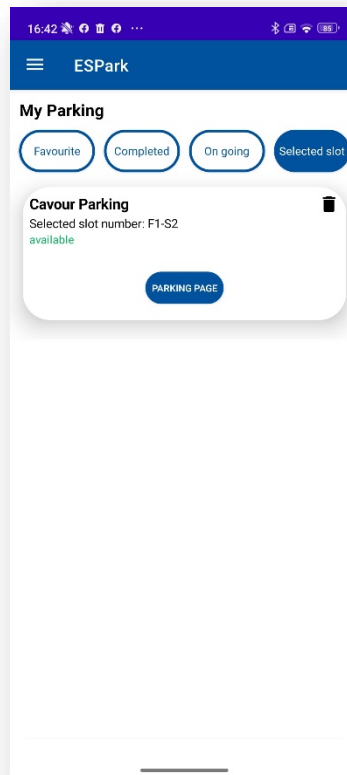
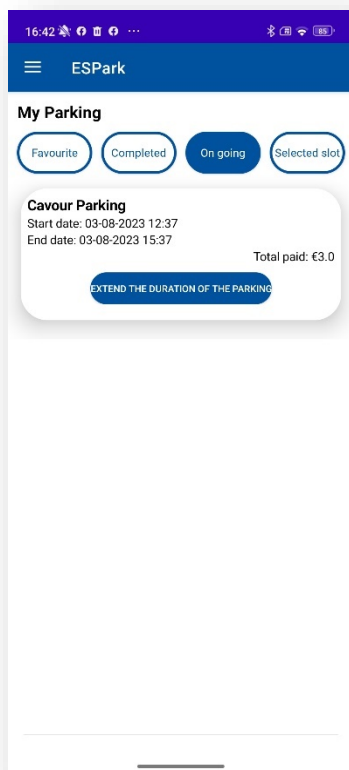
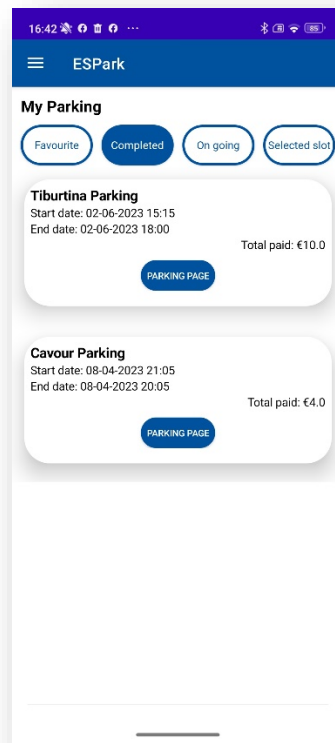
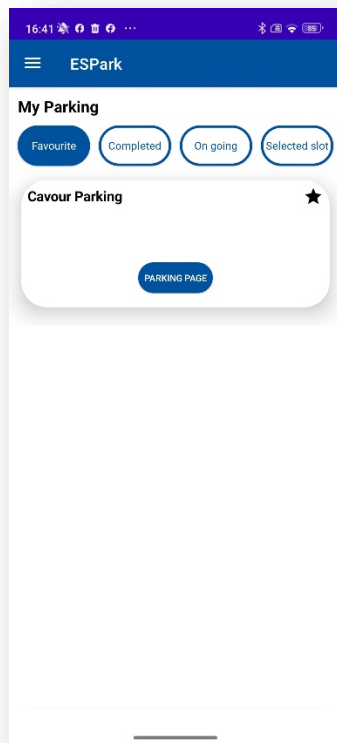
Parking page, map of the parking with all slot and information of “choose an available parking slot”



Payment task with 3 page to insert all info about the payment



Extend parking duration task



My parking page with all info about ongoing, completed and favourite parking, and selected slot to receive info about the availability

Chapter 7

Conclusions and Future Work

7.1 Conclusions

This course and the realization of this project allowed us to use a different approach for the development of an application, that is, having the needs and opinions of the user as the central theme. Compared to other projects, this was the first time that we came into close contact with the user, interviewing him to obtain valuable feedback.

In the end we acquired new skills useful for creating an application, from the point of view of interaction between human and computer.

7.2 Future Work

The future objectives will certainly be to improve the application, adding other interesting features, improving existing features, creating versions for each operating system and adding the possibility of interfacing with the app via voice to avoid distractions while driving.

The idea is also to be able to test our application in the real world, with real parking lots.